

v2405.0.0

CloudStation User Manual.



YellowScan

Designed to innovate.

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SECTION I

Installation.

A solid yellow horizontal line underlining the word 'Installation.'

Installation.

► Minimum Requirements

The minimum requirements for running CloudStation are summarized in the table below, as well as the recommended hardware to run it. CloudStation may require more computing power for specific actions:

- Strip Adjustment and Colorization are using more CPU power
- Classification benefits from the GPU to perform interpolations
- Lastly, Colorization also benefits from additional RAM

	Minimal Configuration	Mid-Range Configuration
CPU	Intel or AMD 4C/8T CPU with SSE4.2 support	Intel or AMD 6C/12T CPU (ex: Intel 8700 or AMD 5600X)
GPU	Intel Integrated Graphics	Dedicated Nvidia GPU with at least 4 GB dedicated VRAM
RAM	8 GB	16 GB
Storage	500 MB for CloudStation Installation	1 TB SSD
Display	1600 x 900	1920 x 1080

► Recommended Configuration



CPU

Intel or AMD CPU

High core count, high frequency
(ex: Intel i9 14900K, AMD Ryzen 9 7950X)



GPU

Dedicated Nvidia GPU

With at least 8 GB dedicated VRAM
(ex: Nvidia RTX 4090)



RAM

At least 1 GB per CPU core, minimum 16 GB.



Storage

2 TB NVMe SSD Gen 4



Display

1920 x 1080 or 4K
(DPI Scaling Enabled)



Using a laptop computer is not recommended as thermal throttling will impact performance from cooling issues.

Installation.

► Setup



YellowScan CloudStation is currently distributed for Windows 10 and Windows 11. The software is not available on Linux nor MacOS.



You can download the latest CloudStation installer [here](#).

Execute the `CloudStation-x.x.x-win64.exe` installer to start the installation. The installation is straightforward, follow the install wizard to the last step. You will have to agree to the End User License Agreement to proceed with the installation.

Once the installation is complete, you can start YellowScan CloudStation by the Desktop shortcut if you ticked the option to create it. Else, you can access the executable in the `\bin` folder. The default installation path is `C:\Program Files\YellowScan\CloudStation`



<https://software.yellowscan-lidar.com/cloudstation/latest/win64>

Installation.

► YellowScan Toolkit



The YellowScan Toolkit (often referred as ytk) is an advanced Command Line Interface (CLI) version of the YellowScan CloudStation software. It allows all processes available in CloudStation with no Graphical User Interface (GUI), with the possibility of being batchable and/or scriptable for all users with a YellowScan CloudStation Ultimate license.

The YellowScan Toolkit installer is bundled with the YellowScan CloudStation installer.

Please refer to the YellowScan Toolkit User Manual for instructions on how to use ytk.



The use of ytk is destined to advanced and experimented users. It contains no guardrail or check on whether something makes sense to do or not in your data. Inconsiderate processing can lead to bad, unusable or incorrect data. User discretion is advised.

First Run Experience.



Activate your license

Select or drag&drop your .key license file here

Browse

► Licensing

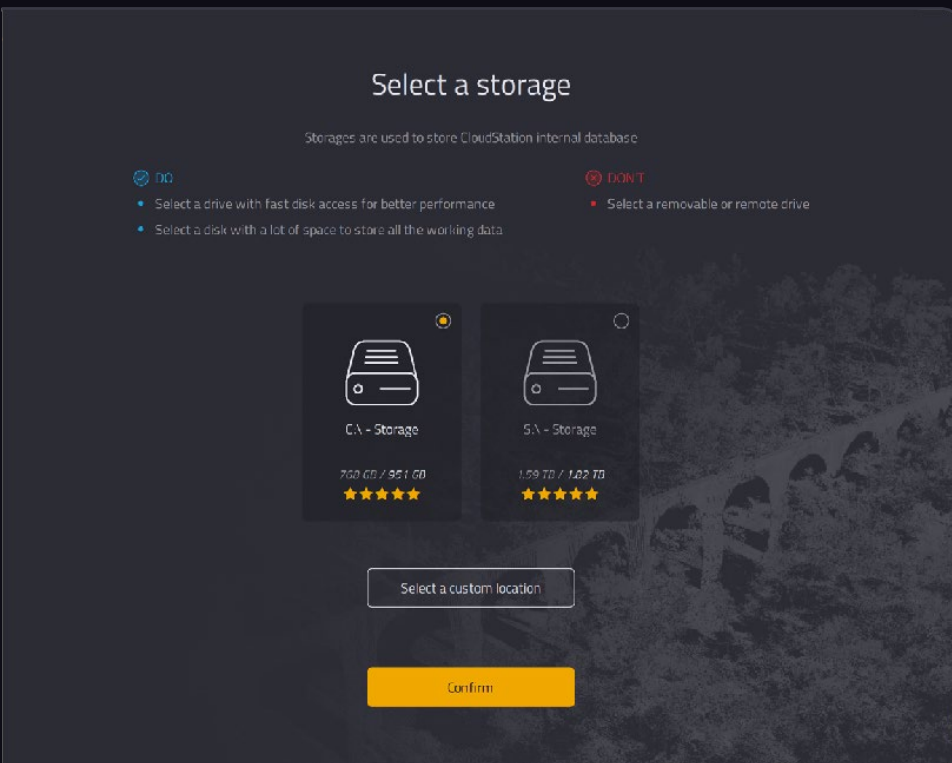
The first time CloudStation opens, a unique window will be displayed, asking for the license file for the software. Before activation, two TCP ports need to be opened in your firewall(s) to be able to access the license server:

- Port 5053
- The second port is unique to your license.

CloudStation will let you know which one to open if needed. Should you have activation issues, refer to the Licensing Troubleshooting section.

Simply drag and drop your .key file or hit the Browse button to navigate to your file.

First Run Experience.



► Storage Selection

The first time CloudStation is ran, or any time no Storage is available, you will be prompted with the Storage selection window.

Each Storage will have a grade based on the hardware used. Ideally, your choice should be on a fast read/write disk, such as NVMe SSD Gen 4.

Once you have chosen your Storage, a folder will be automatically created at its root, named YellowScan and containing all the necessary additional data for each project.

Learn more about how Storages work on [Page 24](#).

First Run Experience.

Select OpenCL device

The device will be used to speed-up some algorithms of the CloudStation



Disable OpenCL

Confirm

► Open CL Device

If your workstation has a graphics card compatible with OpenCL, you can use its computing power to speed up some processes, such as Terrain classification algorithm or MTA resolution on Voyager scanners.

A simple 5-star rating will easily give you an idea of the best GPU to use. We recommend using Nvidia graphics cards (GTX-, RTX- or Quadro-series GPUs) over AMD.

Learn more about OpenCL on [Page 17](#).

Once all steps are complete, you will be landing on the Welcome Screen.

First Start.

Info

See more information about your software here.

Settings

Access the settings from anywhere by clicking on the cogwheel icon here.

Menu Bar

From project creation to storage gestion, it has everything to get you started.

Home Page

This is the first screen you'll see when CloudStation is started.

Projects Filtering & Display

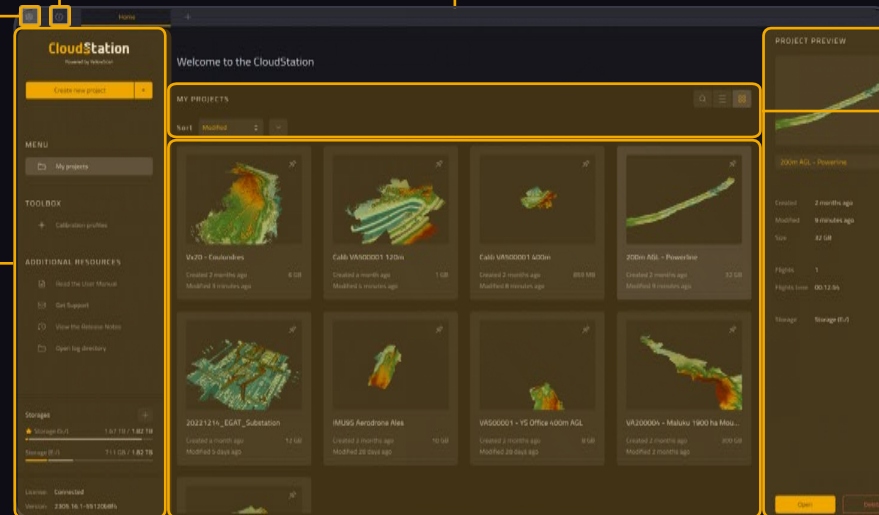
Choose here how to sort and display your projects in your Catalog. You can also search projects quickly with **Ctrl**+**F**.

Project Information

Available when a project is selected. Displays information about your project. Press **Enter** to open.

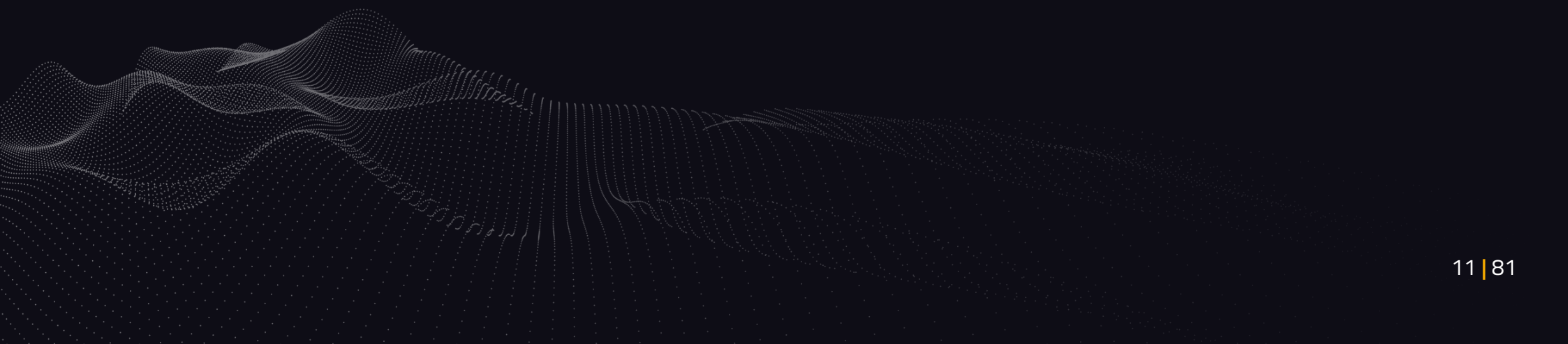
Catalog

You'll find all your projects here.



SECTION II

Settings.

A solid yellow horizontal line positioned under the word 'Settings'.

General Settings.

1

Mouse Warping

When your mouse cursor reaches the side of a viewport during an action, automatically teleport it to the other side to allow continuous movement without hindrance.

2

Subsampling

Tiles the data to optimize visualization performance. When on, a first subsample will be quickly displayed.
(see next page)

3

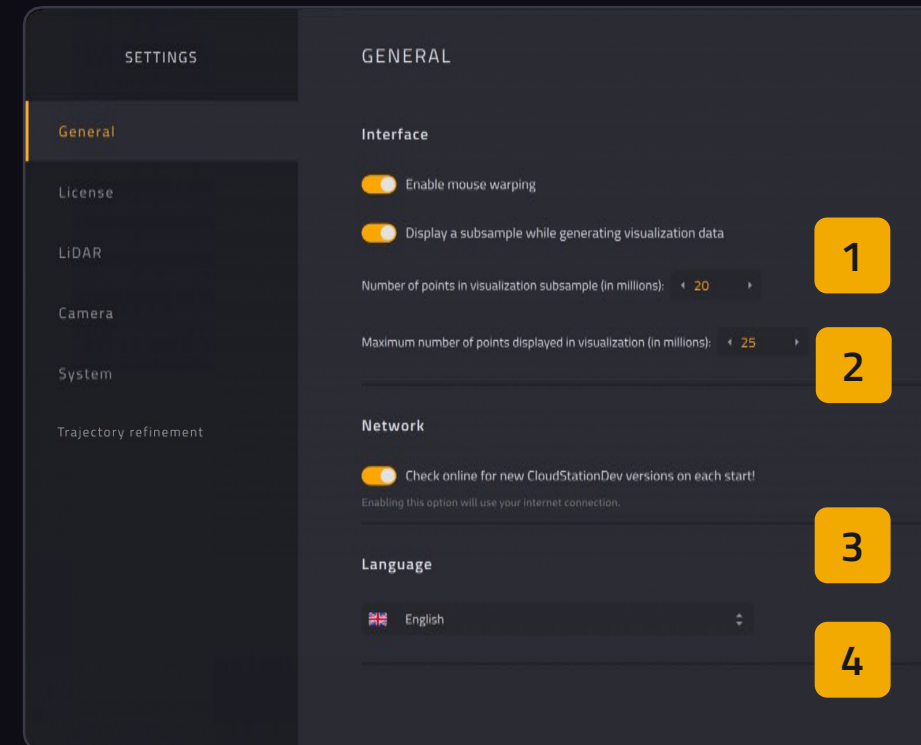
Check for Updates

Automatically check for software updates on startup. This requires an internet connection.

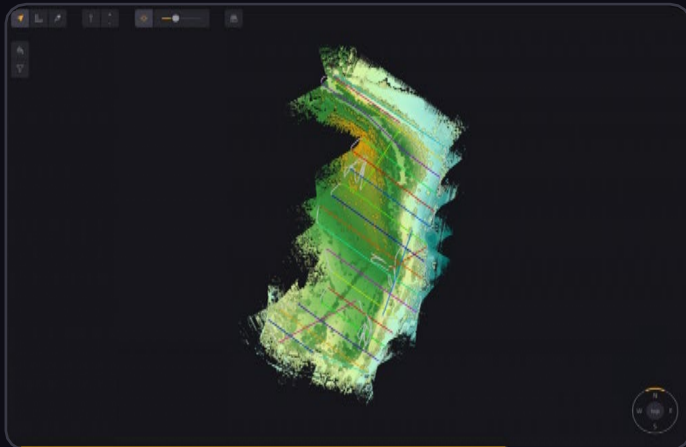
4

Display Language

Choose the language used in text. Currently supported: English, Japanese



General Settings.



► Subsampling ON

Displays a subsample and two-point count options:

CloudStation uses a dynamic subsampling method to display point clouds. This dynamic subsample is long to generate so the CloudStation can display a «regular» subsample during the generation of the dynamic subsample to allow you to see your point cloud faster.

► Subsampling OFF

If you disable the regular subsample, this will make the dynamic subsample generation start sooner.

The two inputs available allow you to choose the number of points in the regular and dynamic subsamples respectively. Default values should work with most GPUs, but it can be beneficial to reduce the values and increase point size on smaller configurations.

► Data Generation Bar

Once the data is extracted, a small bar will be displayed at the bottom of the Viewport indicating the status of both the subsample and the dynamic subsample.

License Settings.

1

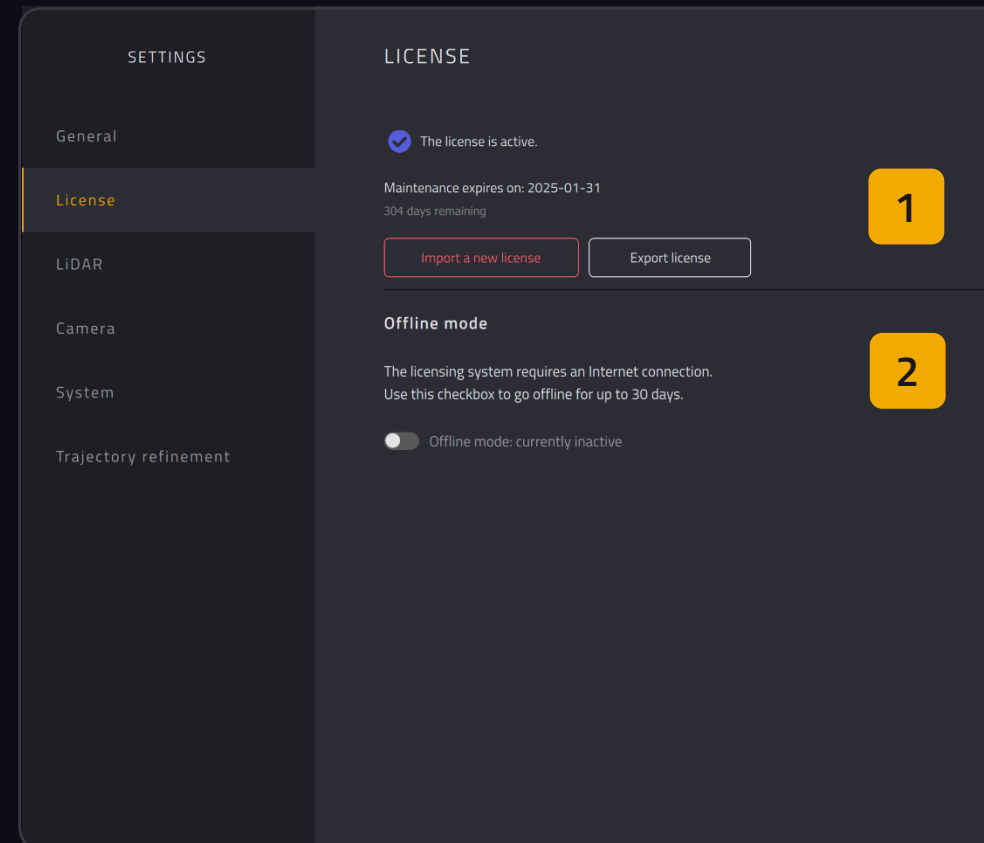
License Status

Displays information about your license and allows you to import new one or export the current one.

2

Offline Mode

Turn ON to use CloudStation without internet access for up to 30 days. Requires an internet connection to activate.



LiDAR Profiles.

1

Profiles List

All your imported LiDAR profiles are displayed here. You can click on the star icon to add one favorite per system type.

2

LiDAR Information

Provides information about your system and its serial number. You can rename your profile here.

3

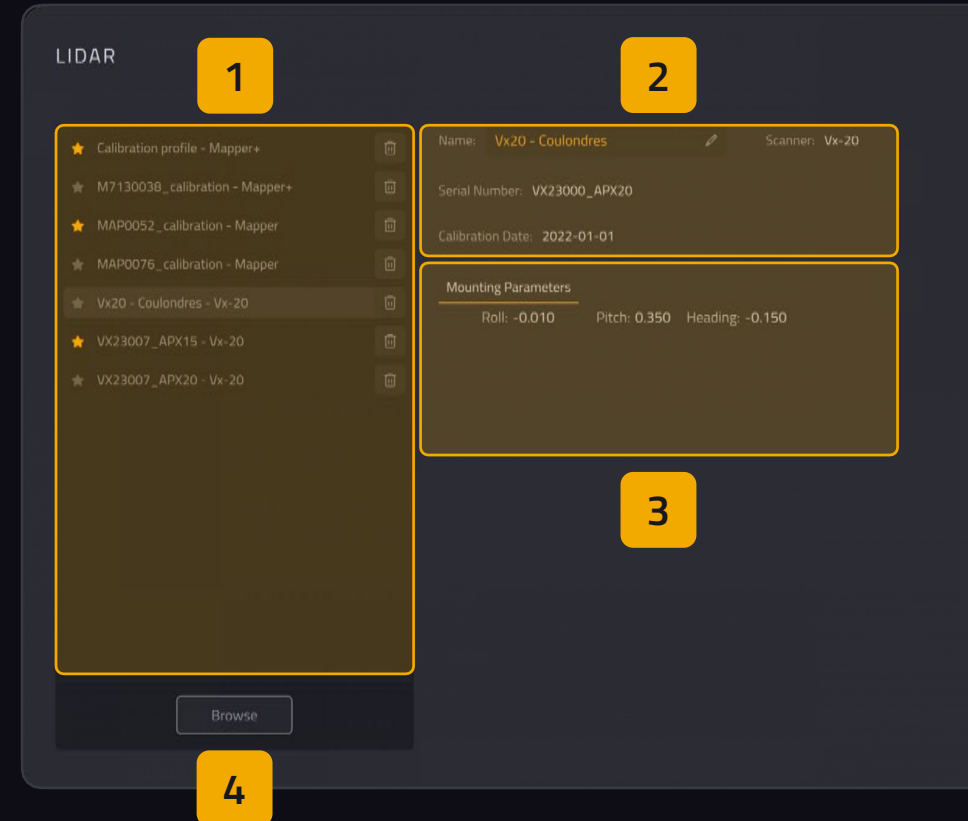
Calibration Method

Calibration used for processing: Mounting Parameters, Per-Beam, Intrinsic or Rislav Intrinsic
(depends on the system)

4

Add a New Profile

You can browse for your **.profile** file here or drag-and-drop it in.



Camera Profiles.

1

Profiles List

All your imported Camera profiles are displayed here. You can click on the star icon to add one favorite per system type.

2

Camera Information

Provides information about your camera and its serial number. You can rename your profile here.

3

Calibration Method

*Calibration used for processing:
Mounting Parameters, Per-Beam or Intrinsic
(depends on the system)*

4

Add a New Profile

You can browse for your **.profile** file here or drag-and-drop it in.

CAMERA

1

2

3

4

Dual Cam - SB400001

Embedded

Single Cam - VA500001

Browse

Name: Dual Cam - SB400001
Scanner Serial Number: SB400001
Serial Number: A0010006
Calibration Date: 2022-12-17

Camera 1

Model: Radial Tangential
Image Width: 5456 Image Height: 3632

Mounting Parameters:

Boresights: Roll: -26.530 Pitch: 0.000 Heading: 90.000

Lever Arms: x: -0.181 y: 0.045 z: 0.000

Intrinsic Parameters:

px: 2731.482 py: 1814.748 s: 0.000 a: 1.000

k1: 0 k2: 0 k3: 0 f: 3757.129

t1: 0 t2: 0

Time offset: 0

Camera 2

Model: Radial Tangential
Image Width: 5456 Image Height: 3632

System Settings.

► OpenCL

If your workstation has a graphics card compatible with OpenCL, you can use its computing power to speed up some processes, such as Terrain classification algorithm or MTA resolution on Voyager scanners.

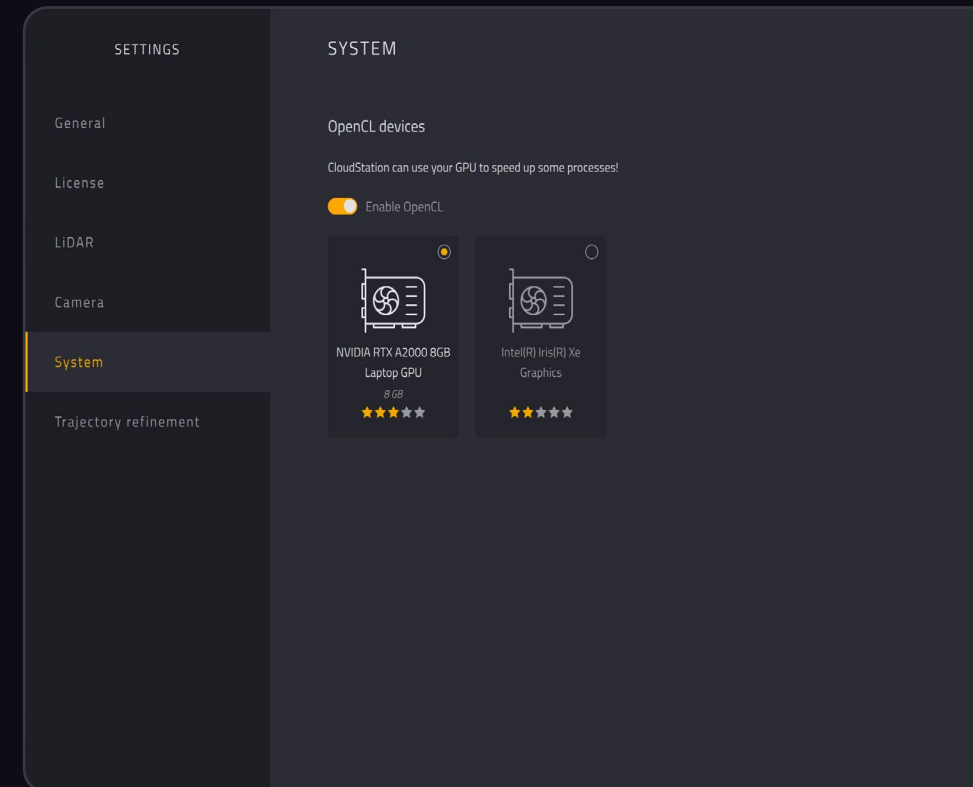


The list of OpenCL compatible devices can be found [here](#).

We recommend using Nvidia graphics cards (GTX-, RTX- or Quadro-series GPUs) over AMD.



When using a laptop, you may have multiple GPUs available, including an integrated graphics one. Typically, you do not want to use integrated graphics over any dedicated GPU, as the latter tends to be more powerful. Refer to the Troubleshooting section to know how to force the use of your dedicated GPU over the integrated graphics.



Trajectory Refinement Settings.

► Configure and Install

CloudStation requires third-party software to refine your trajectory (SBET Generation) and ensure your data is always at the best quality possible.

Depending on the LiDAR system you own, you will have to install either or both of the following:

- Applanix POSPac® UAV or MMS: see [Page 19](#)
- SBG Systems Qinertia®: see [Page 20](#)

Lastly, we recommend always installing [Trimble® ConvertToRinex](#) so CloudStation can import, convert, and process your Trimble GNSS Base Station files during the trajectory refinement process. Its [utilities](#) are also required. Please refer to [Page 21](#) for more information.

Applanix POSPac®

SBG Systems Qinertia®

Trimble® ConvertToRinex

Heads Up

You'll need to know where each software you use is installed! Some third-party software will require administrator rights to be correctly installed.

You will also need to have an active license for each of the software that you intend to use. Licensing for each software is separate from CloudStation.

► Locate Each Software

Locate the .exe file for each software by navigating to the appropriate folder. By default, CloudStation will try to open the default install path for each software.

Applanix POSPac.

1

POSPac Batch Path

Select the installation path where your POSPacBatch.exe executable is located.

2

Optional Licenses

Tick ON the available licenses you have for POSPac to enable them in the SBET generation step.

TRAJECTORY REFINEMENT

CloudStation requires third-party software to refine your trajectories and ensure your data is always at the best quality. Depending on your needs, please configure the appropriate softwares below.

Applanix POSPac® SBG Systems Qinertia® Trimble® ConvertToRinex

PospacBatch.exe path C:/Program Files/Applanix/POSPac MMS 9.0/PospacBatch.exe

1

Enabled licenses

Please select the available licenses in your Applanix POSPac® installation

☐ PP-RTX
☒ SmartBase

2

SBG Systems Qinertia.

1

Qinertia CLI Path

Select the installation path where the **qinertia-cli.exe** executable is located.

2

Login to Your MySBG Account

A login button will open the MySBG account login page. Enter your credentials here. Then, your license information will be displayed in CloudStation. If you have a new key, you can enter it directly in CloudStation.

TRAJECTORY REFINEMENT

CloudStation requires third-party software to refine your trajectories and ensure your data is always at the best quality. Depending on your needs, please configure the appropriate softwares below.

Applanix POSPac® SBG Systems Qinertia® Trimble® ConvertToRinex

qinertia-cli.exe path C:/Program Files/SBG Systems/Qinertia/bin/qinertia-cli.exe

1

Account information

User id Thibaud Capra

License key [REDACTED]

2

Disconnect

Trimble ConvertToRinex.

1

Useful Links

For your convenience, you'll find direct links to both required software.

Also available on [Page 18](#).



About Visual C++

You may need to separately install the x86 Visual C++ Framework for it to work. Find it [here](#).

2

ConvertToRinex Path

Select the installation path where the **convertToRinex.exe** executable is located.

TRAJECTORY REFINEMENT

CloudStation requires third-party software to refine your trajectories and ensure your data is always at the best quality. Depending on your needs, please configure the appropriate softwares below.

Applanix POSPac® SBG Systems Qintertia® Trimble® ConvertToRinex

YellowScan CloudStation can automatically import and process your Trimble GNSS base station files during trajectory refinement. To enable this feature, please download and install Trimble ConvertToRinex.

Trimble ConvertToRinex and its dependencies can be found here: [Trimble ConvertToRinex](#) [Trimble Office Configuration Utility](#)

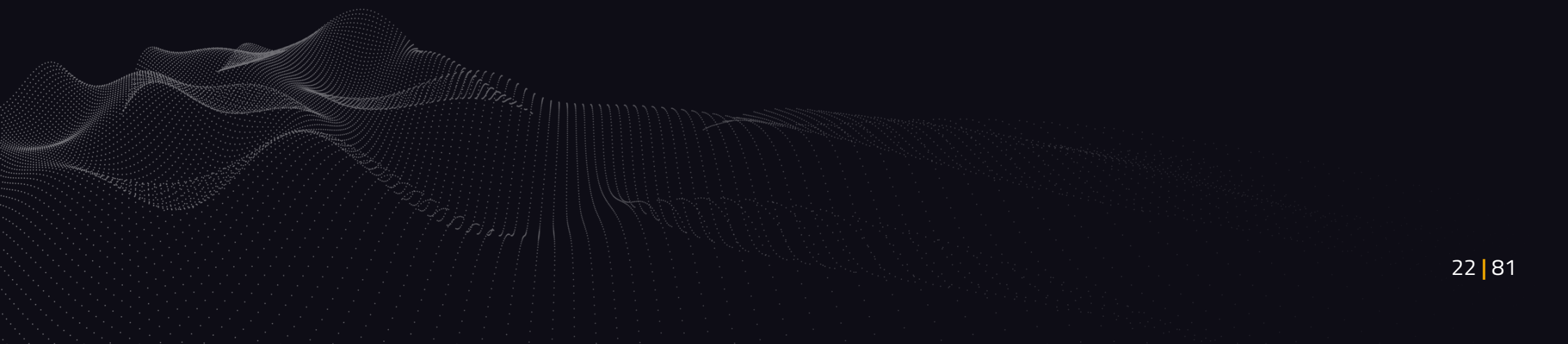
convertToRinex.exe path [C:/Program Files \(x86\)/Trimble/convertToRINEX/convertToRinex.exe](#)

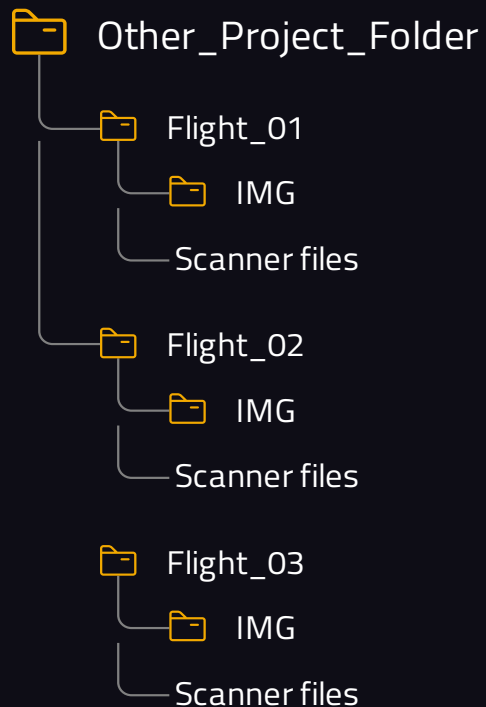
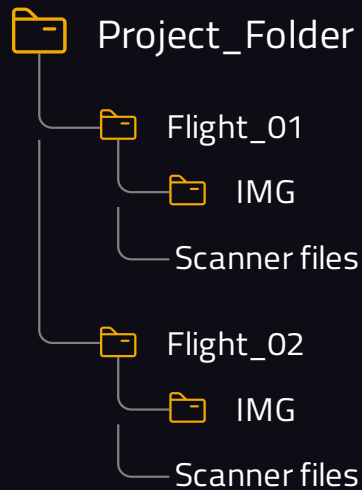
1

2

SECTION III

Workflow.





Organizing your Files.

Quick Guide

Having a clear file organization will save a lot of time when processing, and will allow you to easily return to any of your projects without getting confused or lost. This section will show you how to select your drives and how to easily organize your files.

► Recommended File Organization

We recommend keeping a file organization similar to this:

- Each Project is a main folder
- Each Flight for this project is a subfolder
- Flight folders should contain the images in a subfolder, with the image list
- The laser data should be in the Flight folder



If you collect data from multiple cameras at once (Dual Camera Module, Embedded Camera-systems with any Camera Module), create one separate IMG folder per camera.

CloudStation

Powered by YellowScan

Create new project

MENU

My projects

ADDITIONAL RESOURCES

Read the User Manual

Get Support

View the Release Notes

1

Storages

+

CloudStation Storage.

1

Add a New Storage

CloudStation uses an intermediate representation to work with your projects. Since this is effectively a clone of your acquisition data, it can take a lot of space on your drive.

To prevent cluttering your acquisition data and your personal folders, CloudStation requires that you create a dedicated folder on your drive that CloudStation will use to store intermediate representation of all your projects. This folder is called a «Storage».

You can have up to one storage per physical drive on your computer.

Your projects can then be managed from the Catalog of the CloudStation which will inform you about the space used by the storage on your drive, allow you to move and delete projects, etc.

When first opening CloudStation, you will have no saved Storage. You can either add (+) one right away or you will be prompted to do so when creating your first project.

Storages

+

★ CS Storage 209 GB / 280 GB

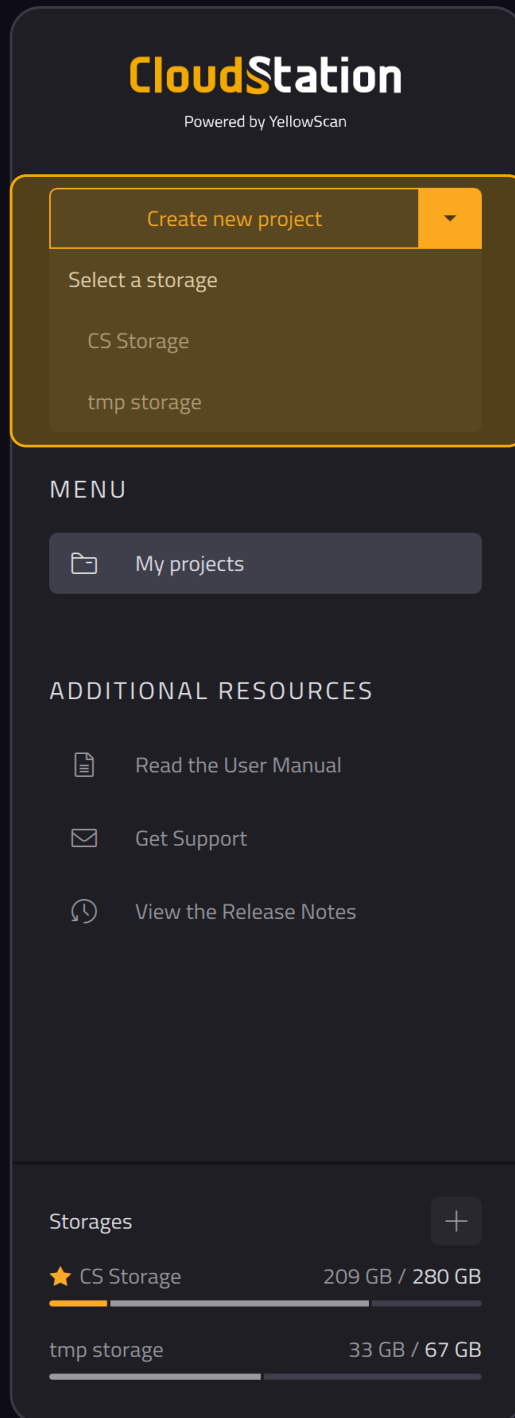
★ △

tmp storage

33 GB / 67 GB

When adding multiple Storages, you can define your favorite Storage for your project to be saved to by default. Hovering a Storage will allow you to add to ☆ Favorite or △ Eject that Storage. By doing so, it will be removed from the list of available Storages.

2



1

CloudStation Storage.

2 Use a Different Storage

If you do not want to use the favorite Storage and have multiple Storages saved, you can select which one to use for this project creation.

If you do not select a specific Storage or only have one Storage defined, CloudStation will use the default one, which is your favorite in the case of multiple Storages available.



CloudStation algorithms benefit greatly from fast disk access, to get the best performance out of the CloudStation, you should create storages on very fast drives (SSDs) with a lot of space to store all the working data.

Workflows & Licenses.

This User Manual aims at describing all steps available in CloudStation.

The overall workflow in CloudStation can be split in 4 different categories:

- | | | | |
|------|-----------------|-----|-----------------|
| I. | Data Generation | II. | Data Refinement |
| III. | Data Enrichment | IV. | Data Export |

However, depending on your CloudStation license, you may or may not have access to a specific feature. License-restricted features are marked with the minimal license necessary to access it. The following tags will be present every time a minimal license is required:

Essential Requires CloudStation **Essential** or higher.

Pro Requires CloudStation **Pro** or higher.

Ultimate Requires CloudStation **Ultimate**. There is nothing higher.

WORKFLOW

Data Generation.

A solid yellow horizontal bar.

Create new project

Select a storage

CS Storage

tmp storage

MENU

My projects

ADDITIONAL RESOURCES

Read the User Manual

Get Support

View the Release Notes

Storages



★ CS Storage 209 GB / 280 GB

tmp storage 33 GB / 67 GB

Data Generation.

► New Project Creation

To create a new project, click on the Create new project button and browse for your Project Folder. CloudStation will automatically search the selected folder for all flights available.

A new window opens, and you can select the flights to import. It displays information regarding the scanner type and date of survey.

INITIALIZE A PROJECT

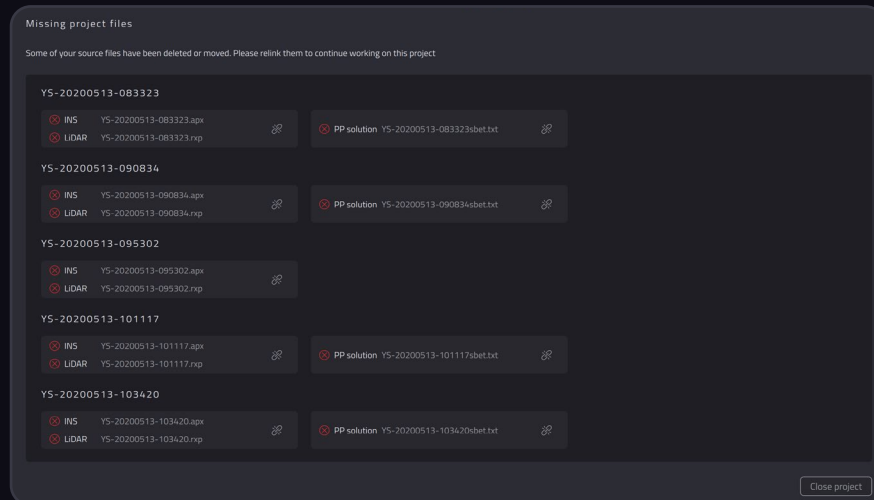
Select the INS files you want to add to your project

☒	Unselect all			
☒	YS-20220427-100314	...res/Golf-PicStLoup/YS-20220427-100314	2022-04-27 10:03:14	Vx-20
☒	YS-20220427-104911	...res/Golf-PicStLoup/YS-20220427-104911	2022-04-27 10:49:11	Vx-20
☒	YS-20220427-111921	...res/Golf-PicStLoup/YS-20220427-111921	2022-04-27 11:19:21	Vx-20

Cancel

Create

Data Generation.



► Lost Project Opening

If the source files (INS, LiDAR, PP solution) of an existing project cannot be found by the CloudStation, it will prompt you with a window that lists all the resources that need to be found.

For each missing resource, point the CloudStation to its new location. Once all resources have been found, the CloudStation will open the project.

Main Window.

Module Selection

Contains all your modules. Click on the module's icon while active to hide the Main Sidebar and maximize the Viewport size. A small check mark (✓) will let you know if you've already used a module on this dataset. Selecting a module will change the Main Sidebar accordingly. Clicking the active module minimizes the Main Sidebar.

Main Sidebar

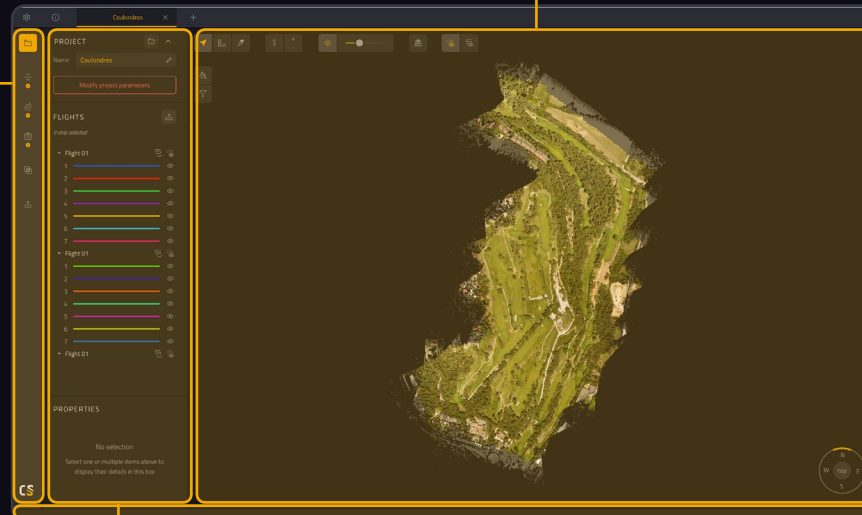
This area grants you access to all the settings for your active module. Refer to each module's section to learn more about their settings. This bar changes depending on the active module.

Viewport

This area displays the trajectory and the point cloud once it's extracted. The floating buttons will give you access to numerous tools.

Shortcut Contextual Bar

Displays contextual shortcuts depending on where your mouse cursor is located, and what actions are available.



Main Window.

► Viewport

This is where your trajectories and point cloud will be displayed.

1

Visualization Tools

This bar regroups tools to manipulate and visualize the extracted point cloud

2

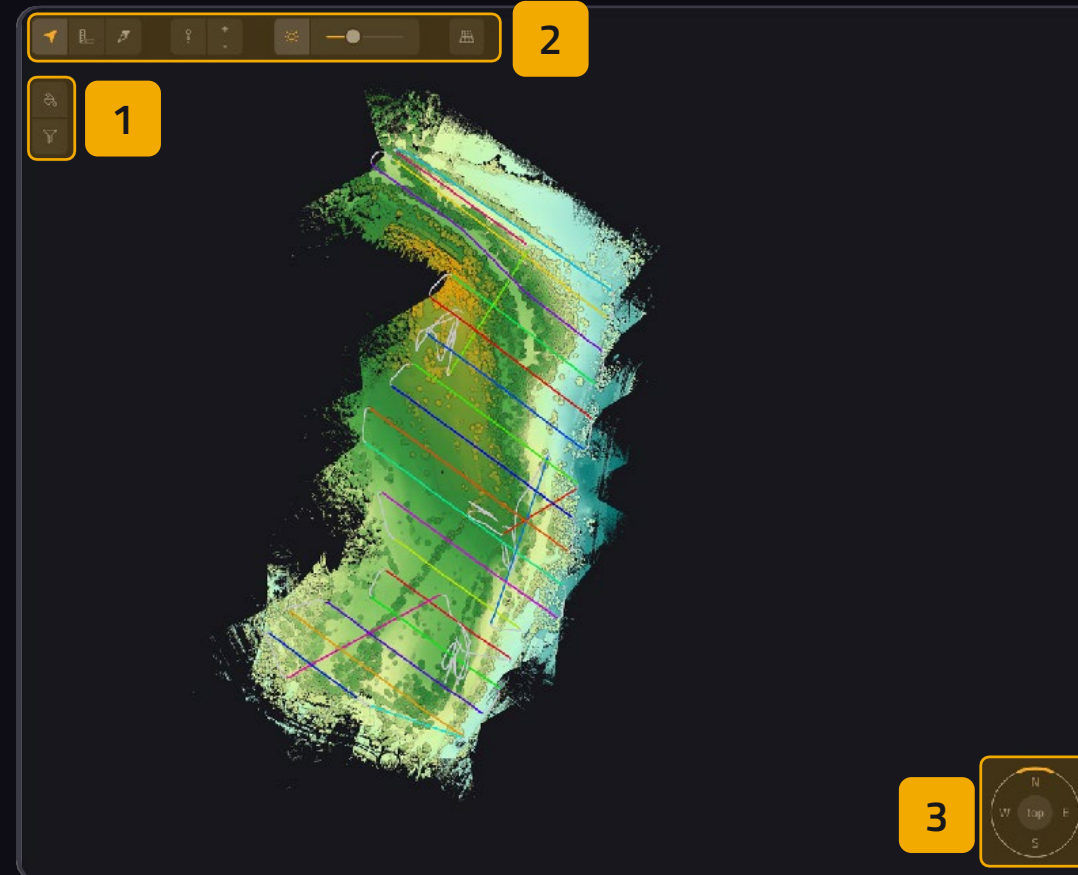
Point Cloud Color & Filtering

These two icons open separate contextual sidebars to change the point cloud color and filter the points displayed based on various attributes and values.

3

Compass

Display the current orientation of the point cloud. Click on a cardinal point to quickly orient the point cloud .



Viewport.

► Tools Available Before Extraction

These tools will help you navigate, measure, and edit your Strips.



Navigation Mode

Select this tool to navigate freely through the point cloud



Measure Tool

Select this tool to get the X, Y, Z coordinates of a point in a chosen projection or to measure distance between two points.



Orthometric & Perspective Modes

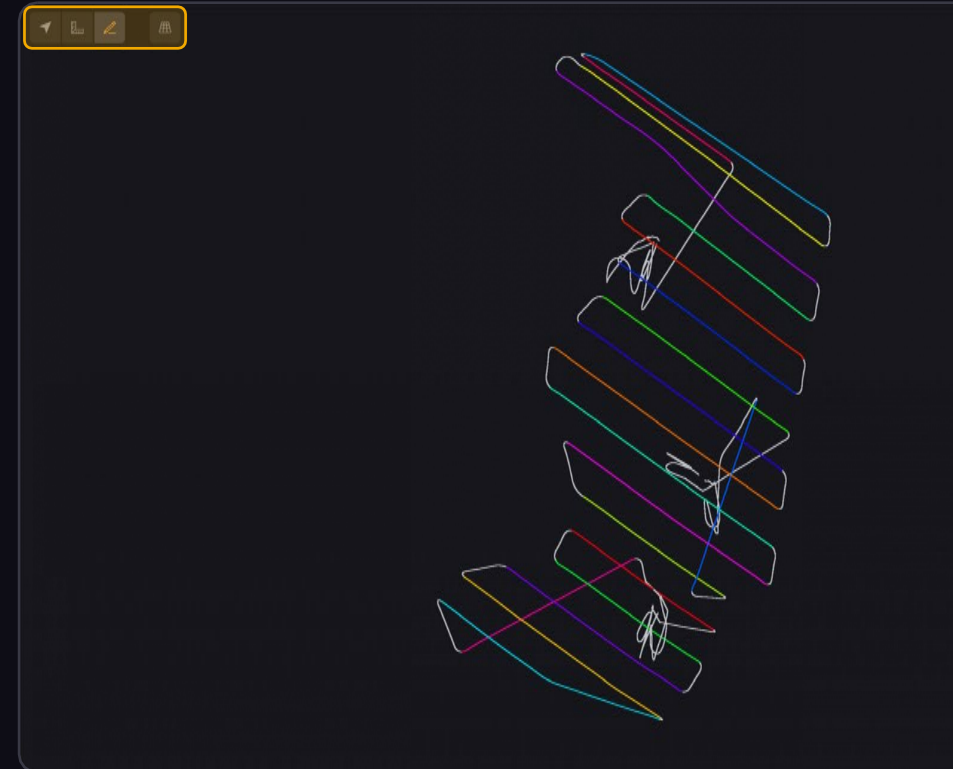
Switch between two different viewing modes. Press **P** to quickly switch



Edit Strips

Before extraction, Edit Strips will allow you to select and edit Strips.

(see [Page 38](#) for more info)



Viewport.

► Tools Available After Extraction

These tools will help you navigate, measure, and edit your point cloud.



Navigation Mode

Select this tool to navigate freely through the point cloud



Measure Tool

Select this tool to get the X, Y, Z coordinates of a point in a chosen projection or to measure distance between two points.



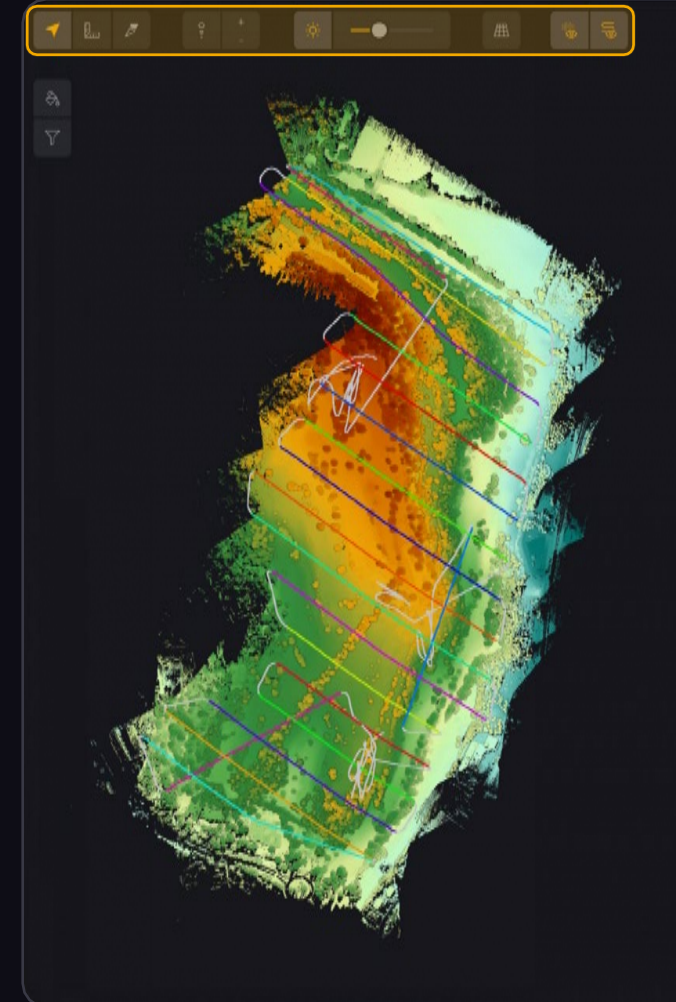
Slicing Tool

Create vertical sections in just two clicks.



Point Size

Change the size of the points in the Viewport



Viewport.

► Tools Available After Extraction

These tools will help you navigate, measure, and edit your point cloud.



Eye-Dome Lighting

Shading technique improving depth perception. Adjust the intensity with the slide bar or turn it ON/OFF by clicking the icon. Often called *EDL*.



Orthometric & Perspective Modes

Switch between two different viewing modes. Press **P** to quickly switch modes.



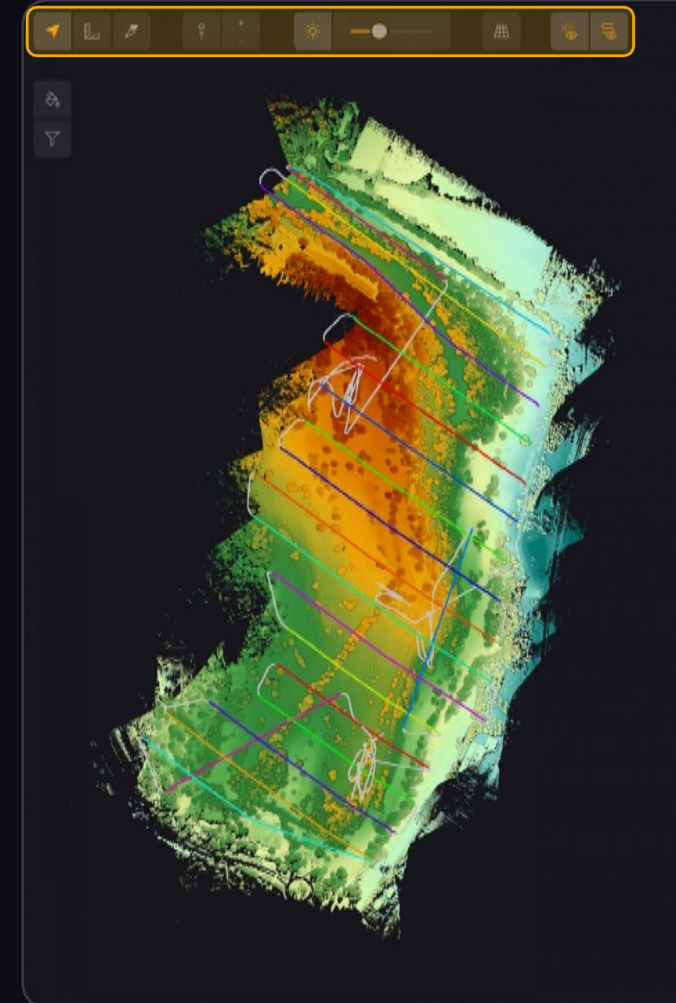
Trajectory Display Toggle

Click to show or hide all trajectories at once in the Viewport

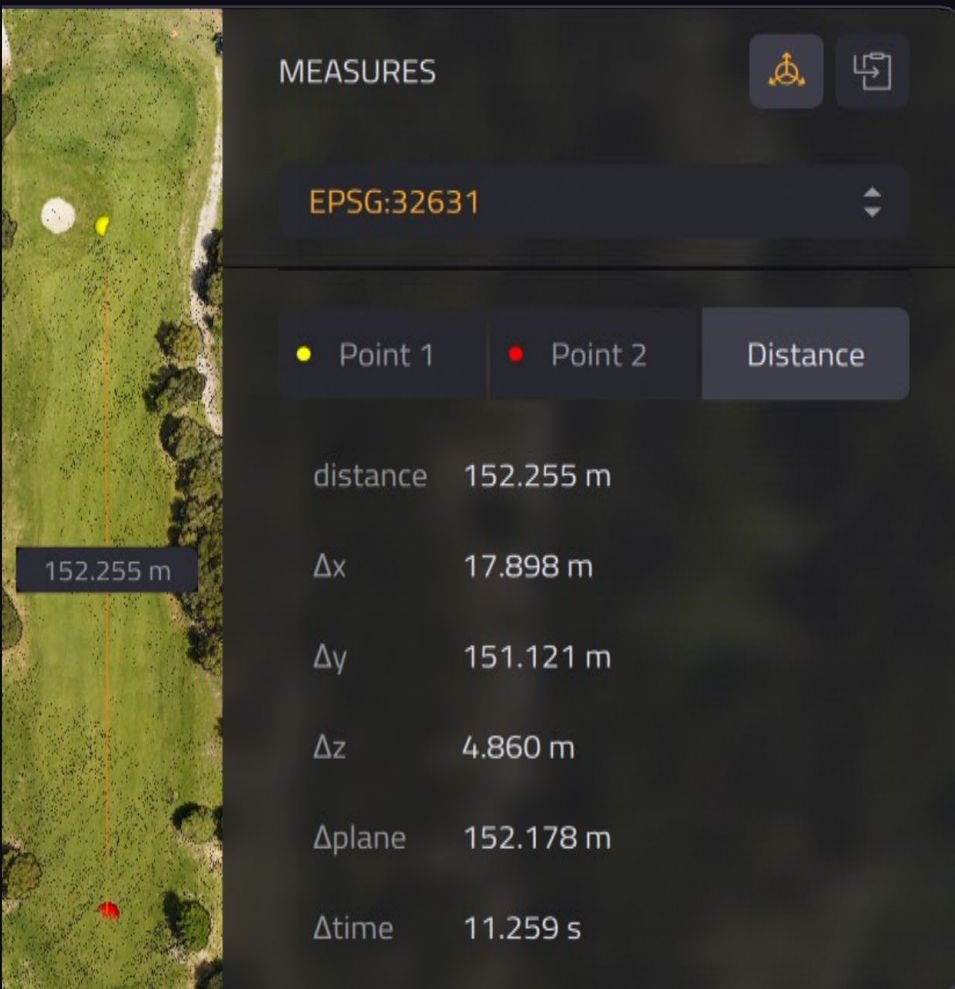


Point Cloud Display Toggle

Click to show or hide all point clouds at once in the Viewport



Measure Tool.



Quick Guide

The Measure Tool allows you get the coordinates of two points in a coordinate system as well as information relative to the distance between them. Simply click the first and last point!



Select a Coordinate System

Choose the CRS in which points coordinates will be given.



Copy to Clipboard

Copies all the point and distance information in your clipboard.

► Point 1 and Point 2

Displays the information about the respective point.

► Distance

Displays information relative to the distance between the two selected points. The value shown in the viewport corresponds to the 3D distance between the two points. For the horizontal distance, see Δ_{plane} .

Slices.



► Slicing tool

When using the Slicing tool, you enter the section limits in two clicks: slice start is symbolized with a green sphere, slice end with a red sphere.

► Slice view

After you clicked the end point, an additional view panel appears at the bottom of the viewport, displaying a 0.20m-wide vertical section by default. You can adjust it using the <|> Slice Width box.

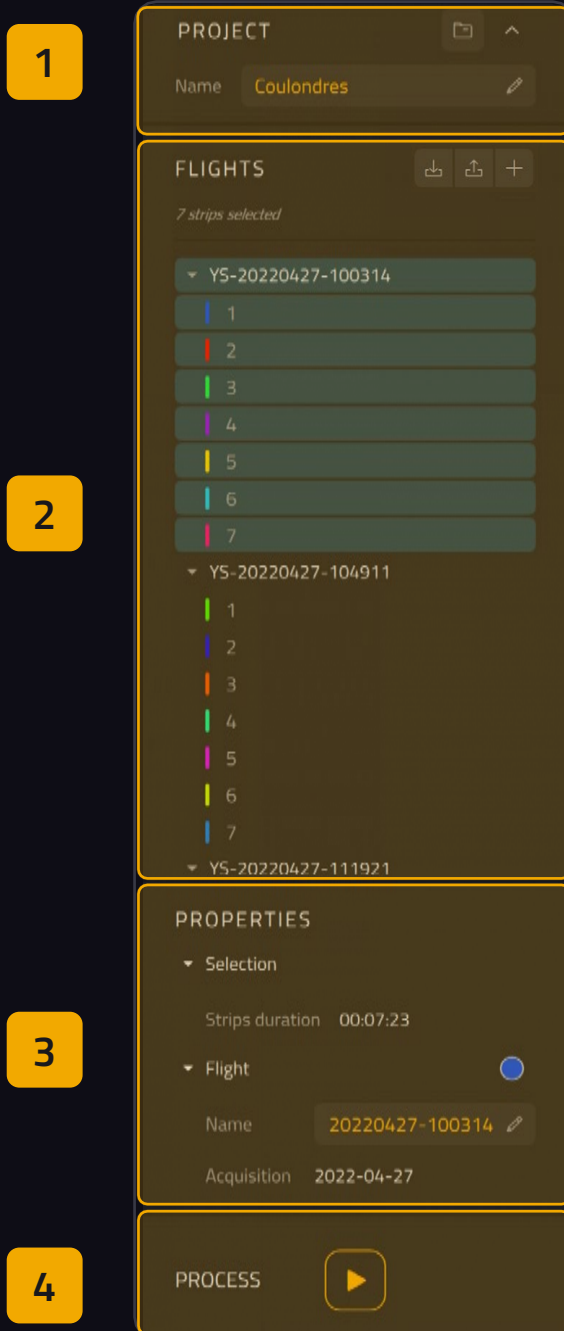
<|> 1.00 m

► Multiple Slices

You can have multiple Slices displayed in the Slice view by creating new ones without closing it. To center the view on a specific Slice, select it in the dropdown list. You can Show or Hide specific Slices, Rename and Delete them (*when hovering*). If you have no Slice left, the view will close automatically.



Accidentally rotated a Slice? Press to reset the view!



Data Extraction.

► Main Sidebar

Prepare the extraction and handle your project there.

1

Project Name

You can rename your project here. This will be the name displayed in the Catalog.

2

Flight and Strip Edition

Your Flights and their respective Strips will be displayed in this section.

3

Properties

Displays information about your current Flight and Strip selection. Selecting a single flight allows you to rename it, while multiple flights will display the total number of points.

4

Process Button

Click here to process the data and extract the point cloud.



Open Project Directory

Access your Project's directory easily with this button.



Add New Flights

Forgot a few Flights? You can add more with this button.

Data Extraction.

► Selecting Strips

CloudStation offers two different ways to work with Strips:

- **Automatic Strip selection:** when loading a trajectory for the first time, CloudStation automatically suggests Strips based on the complete trajectory analysis.
- **Manual Strip edition:** once the trajectory is loaded, you can edit any Strip by clicking on  Edit Strip in the Main Sidebar or directly clicking on the Strip in the Viewport.

This brings a slider at the bottom of the Viewer where you can extend or trim the strip timing by moving the cursor. Press **Enter** or click the Save button or at the end of the slider to validate your change. Press **Esc** to cancel and exit edition. Navigate the slider by using  to pan, and  to Zoom In / Zoom Out.

When editing a Strip, other Flights will be heavily dimmed, while the Strips from the same Flight will be greyed out. Your active Strip is highlighted by its color.

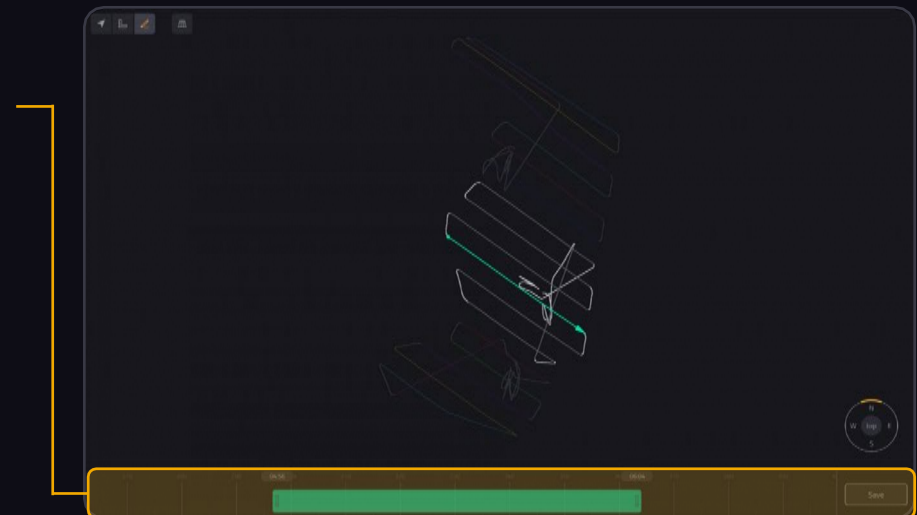


You can also  Import and  Export Strip timestamps to share them with others and have the same data.

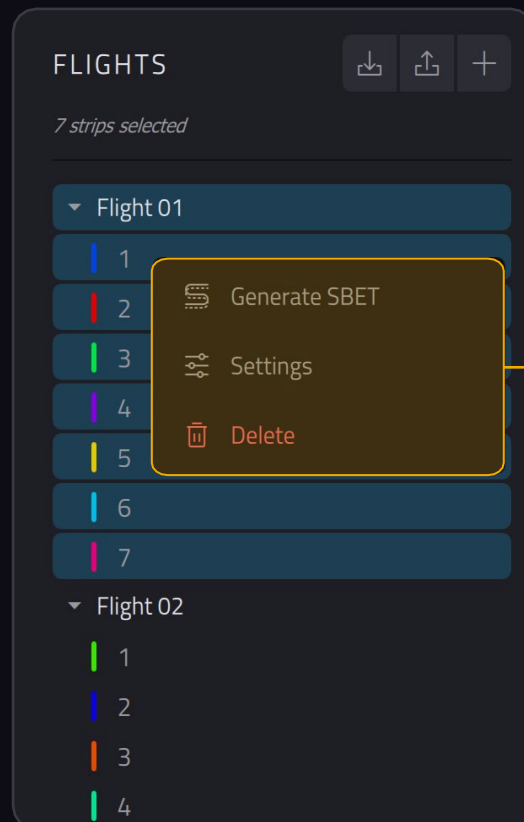
Quick Guide

Before manipulating or running any sort of algorithm, the point cloud needs to be extracted by selecting Strips. It's the first step towards your final point cloud!

Select your Strips, define each Flight's settings and run the extraction to get your point cloud.



Data Extraction.



► Handling Multiple Flights

When working with multiple Flights, you can select an entire Flight (and all its Strips) by clicking on the Flight's name. To select multiple Flights and/or Strips, use **Ctrl** + . To directly select a range of Flights and/or Strips, use + to define the range. You can then access multiple actions with your selection:

Contextual Menu

Access the Contextual Menu with



Start SBET Generation Process

Opens the pop-up window to generate SBETs for your current Flight selection.



Edit the Current Selection

Opens a new pop-up to edit the Flight settings. You can also bulk edit all the Strips selected. (See next page)

This menu is also available for individual Flights and/or Strips by clicking on the Edit icon when hovering a given Flight or Strip.



Properties

Delete your current selection (Click and hold).
This action cannot be undone.

Data Extraction.

► Flight Settings

FLIGHT SETTINGS 1 flight

☐ PP Sol...

LiDAR

VX23009_APX20

Strip Detection

◀ 328.00 m ▶

STRIPS SETTINGS 7 strips

Angle Range

◀ -60 ° ▶

◀ 60 ° ▶

Min Laser Range

◀ 20.00 m ▶

Max Laser Range

◀ 700.00 m ▶

Outlier removal

Turn Off

- **PP Solution:** When you have only one Flight selected, you can select the Post-Processed Solution (SBET). If you already have it in your Laser folder when importing the data, the SBET will automatically be detected and selected. You can still override the SBET by generating a new one. Note: This is unavailable if you have multiple flights selected.
- **LiDAR:** Select the LiDAR profile for your selection.
- **Strip Detection:** Sets the minimum length for CloudStation to attempt to detect Strips automatically. Changing the value will immediately replace your current Strips by the ones newly detected.



Mismatched parameters

When flights or strips in your selection have different values for the same attribute (ie: scan angles), they enter a “mismatched” state, characterized by 3 dots in the input fields.

To override the value for all items in your mismatched selection, click on the field and type the new value.

Angle Range

◀ ... ▶

◀ ... ▶

Data Extraction.

► Strip Settings

FLIGHT SETTINGS 1 flight

☐ PP Sol...

LiDAR VX23009_APX20

Strip Detection 328.00 m

STRIPS SETTINGS 7 strips

Angle Range -60 ° 60 °

Min Laser Range 20.00 m

Max Laser Range 700.00 m

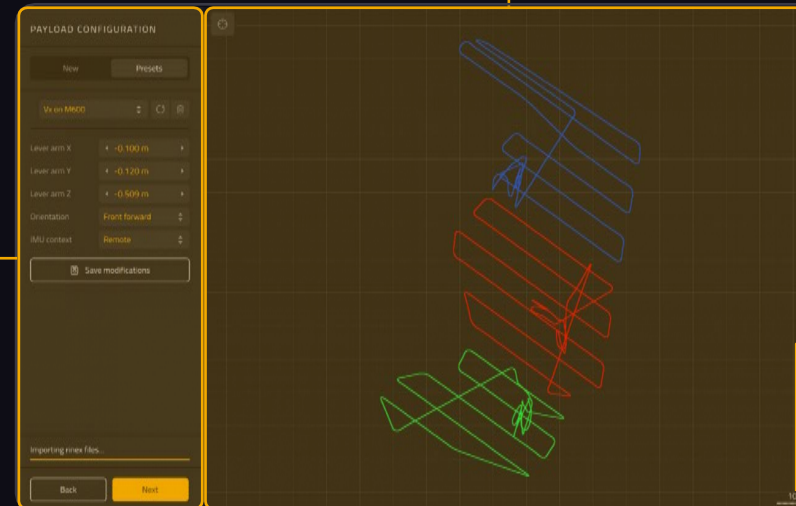
Outlier removal Turn Off

- **Angle Range:** Sets the opening angle to extract data. Some systems have a set Field of View, so the Angle Range will be capped.
Note: The angle range is displayed Front Forward.
- **Min. Laser Range:** Smallest distance from the LiDAR to generate a point. Can be useful to filter potential landing gear.
Default: 1.00m.
- **Max. Laser Range:** Longest distance from the LiDAR to generate a point. Can be useful to filter potential outliers.
Default: 10000.00m.
- **Remove Outliers:** Attempts to remove outliers from the generated point cloud. Note that even after the application of the filter, few outliers may remain. You should usually leave this ON. However, this can remove points on powerlines if the density is too low.

SBET Generation Process.

Main Bar

On this side of the window will be where most of your actions will be completed, varying at each processing step.



SBET Generation Pop-Up

This is where all the steps to generate your SBETs will be completed.

Scale

The bar represents one main square, divided in tenths.

Trajectories Display

All processed Trajectories and Base Stations will be displayed here.



You can generate multiple SBETs at once as long as all of your selected Flights have the same mounting configuration.

PAYLOAD CONFIGURATION

New

Presets

Vx on MG00

Lever arm X -0.100 m

Lever arm Y -0.120 m

Lever arm Z -0.509 m

Orientation Front forward

IMU context Remote

Save modifications

Importing rinex files...

Back

Next

Payload Configuration.

Quick Guide

The SBET Generation process will guide you step-by-step through a simplified workflow to get your trajectories refined before extracting your data. Always check your data quality with the QC Reports!

1 Payload Presets

If you have a Preset saved, you will land on the Presets page. Else, you can save a Preset once you've entered a new one and reuse it in future Projects.

2 Lever Arms Settings

Enter the Lever Arms for your configuration according to your IMU Reference Frame and choose your system orientation if necessary. By default, your system is « Front Forward ». Entering a name for a New Preset allows you to save it and reuse it in future Projects.

3 Automatic Rinex Import

Rinex files found in all Project folders will automatically be imported at this step to be used later. You will be able to add any missing ones on the next step.



Accepted rinex formats: *.T02, *.??o, *.??d, *.rnex, *.crx.

Base Station Configuration & Attribution.

Base Handling
Import, split, edit and configure
all your Bases here.
More details on the next page.



Flights/Base Attribution

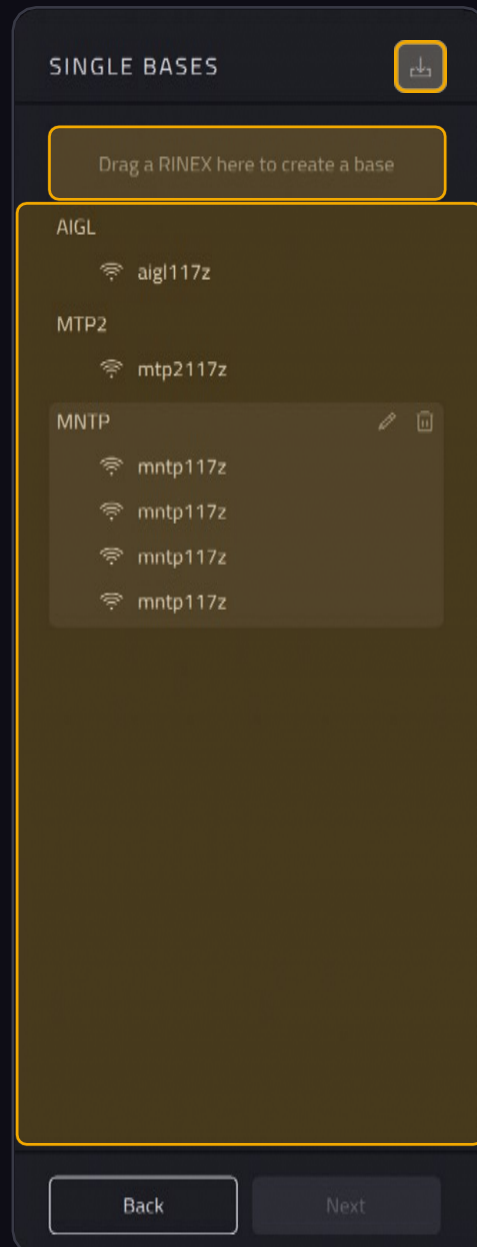
We automatically attribute the closest overlapping Base Station to each Flight.

You can manually edit this selection for other methods or Base Station. If no overlap is available, you will have a warning sign next to the flight. Manually edited attributions are in yellow. Other methods will require the proper license to work:

- Applanix SmartBase™
- Trimble PP-RTX®

Trajectories Display

All processed Trajectories and Base Stations will be displayed here.



Single Base Configuration.

► Handling Base Stations

1 Import Rinex Files

If you have a Preset saved, you will land on the Presets page. Else, you can save a Preset once you've entered a new one and reuse it in future Projects.

2 Base Settings

By default, your rinex files are grouped by Base name if they have the same antenna height. You can drag individual rinex to separate them or create a new Base. Click on  Edit to access the Base Coordinates Settings. Click on  Delete to remove a Base. This cannot be undone.

Edit Base Coordinates.

EDIT BASE

×

Modify current base station parameters

📷

Base name

MNTP

✎

Coordinates

Reference frame

Antenna parameters

Latitude

43.646233558589 °

Frame

ITRF2014

↕

Height

0.0000 m

▶

Longitude

3.869116404786 °

Ellipsoid

GRS_1980

↕

Reference

Antenna reference point

↕

Altitude

104.0362 m

Epoch

Mission date

↕

Apply a preset

Base name	Frame	Ellipsoid	Epoch	Latitude	Longitude	Altitude	
MNTP	ITRF2014	GRS_1980	Mission Date	43.646233558589°	3.869116404786°	104.0362 m	🗑

Ok

► Base Coordinates

Enter your Base coordinates here if it's on a known point. Do not forget your Reference Frame! If you often use this Base, you can save its coordinates for future use by clicking on 📷 Save in the top right corner.

► Saved Bases

All your previously saved Bases will be displayed here. Select a Base in the list to apply its saved coordinates to your current Base. Saving a Base with an existing name will overwrite it.

Click on 🗑 Delete to remove it from the list. This cannot be undone.

2

EXPORT PARAMETERS

1

Height options

Height option **Orthometric**

Geoid model **RAF 18 (France)**

2

Mapping frame

Frame **ETRF2014**

Target epoch **Custom date**

2010-01-01

3

Back Process SBET

Export Settings & Process.

1

Height Options

If you have a Preset saved, you will land on the Presets page. Else, you can save a Preset once you've entered a new one and reuse it in future Projects.

2

Mapping Frame

Define your Mapping Frame settings here. We default to Mission Epoch but you can enter a custom date as well

3

Process SBET

When everything is ready, hit the Process SBET button to start the generation. Once finished, a summary of all processes will be displayed in the main bar.

Flight 3

Importing POS data



If the results are satisfactory, press Apply SBET to automatically assign each successfully generated SBET to their respective flight. Else, click and hold the Cancel button to abort processing.

☒ Flight 1
 ☒

☒ Flight 2
 ☒

☒ POSDataImport
 ☒

☒ BaseStationImport
 ☒

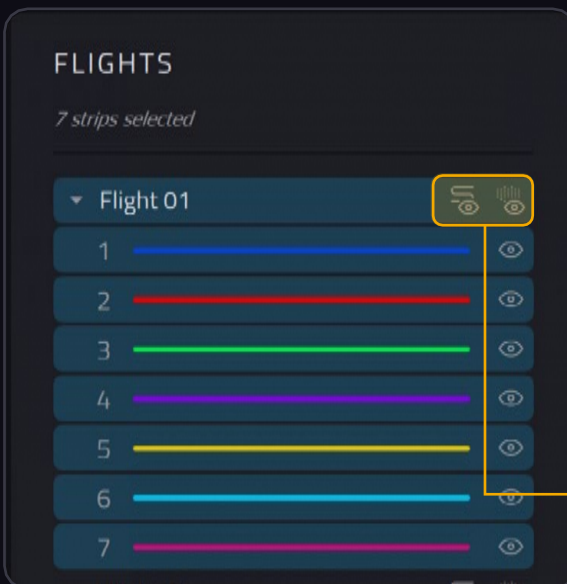
☒ GNSSInertial
 ☒

☒ Export
 ☒

☒ QcReport
 ☒

☒ Flight 3
 ☒

Simple Data Extraction.



Main Button

Accepted rinex formats: *.T02, *.??o, *.??d, *.rnx, *.crx.

► Processing Progress

You can follow the processing progress in different locations. Each Strip will turn into an individual progress bar. The Main Button will have a bar circling around it, showing the global progress. Lastly, your Windows taskbar will also display the global progress as a green overlaid bar on the CloudStation icon. Once complete, your resulting point cloud will be displayed in the Viewport.

► Displaying Data and Trajectories

You can handle the display of data and trajectories in the Viewport by using these buttons while having Flights and/or Strips selected



Show/Hide Trajectories

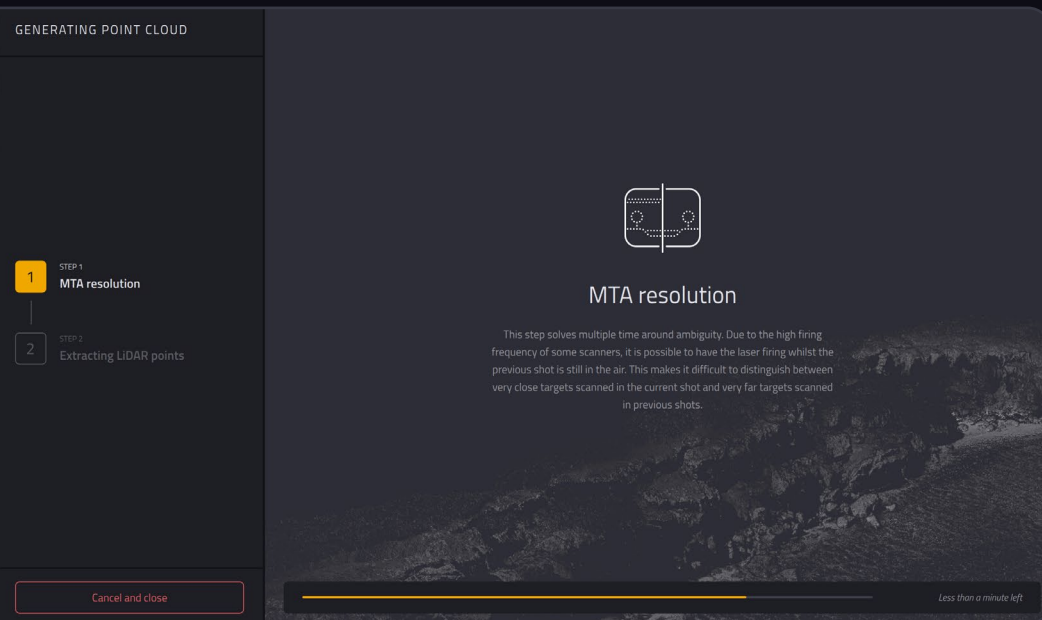
Toggles the visibility of trajectories in the Viewport.



Show/Hide Point Cloud

Toggles the visibility of the point cloud from a Flight and/or Strip.

Advanced Data Extraction.



About

This section describes the specific extraction process for YellowScan Voyager and Navigator systems. Other systems will use the Simple Data Extraction. The steps in the Advanced Data Extraction process may vary depending on the data being processed.

► Voyager Advanced Extraction

When extracting the data from any Voyager flight for the first time, an additional progress bar will show, displaying the progress of the MTA solving for the flights.

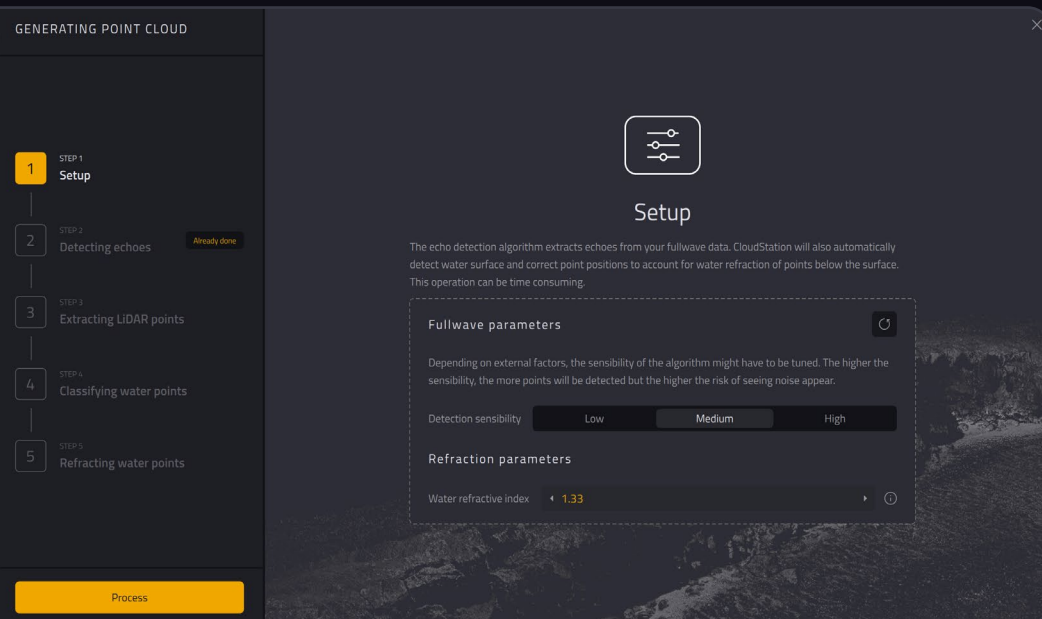
This progress bar will only show on your first extraction. If you modify your project parameters later, this step will already be completed and saved, so you will not have to go through it again.



Solving MTAs can be a very lengthy process, depending on your flight conditions and MTA zone.

Once the MTAs are solved, you will move on to the Extraction step, similar to the Simple Data Extraction.

Advanced Extraction Window.



► Navigator Advanced Extraction

To generate accurate data from any Navigator flight, multiple steps are required to extract it. The Advanced Extraction Window will guide your step by step to go through Extraction Setup, Echo Detection, LiDAR extraction and the Iterative Water Classification to properly apply the water refraction.

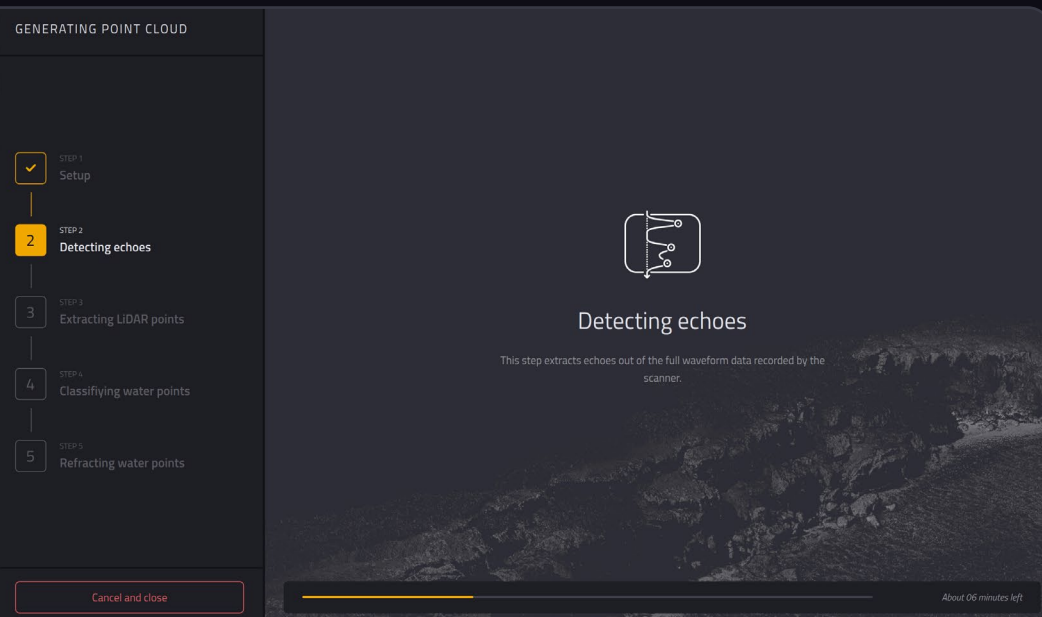
1

Setup

Select the extraction parameters during this step:

- **Detection Sensibility:** Select your echo detection threshold. Increasing the sensibility may result in better water penetration but may add noise to the data.
- **Water Refraction Index:** Select the value for the water refraction. The default is 1.33, corresponding to fresh water.

Advanced Extraction Window.



► Navigator Advanced Extraction

2

Echoes Detection

Detection of the echoes from the full waveform data happens during this step..



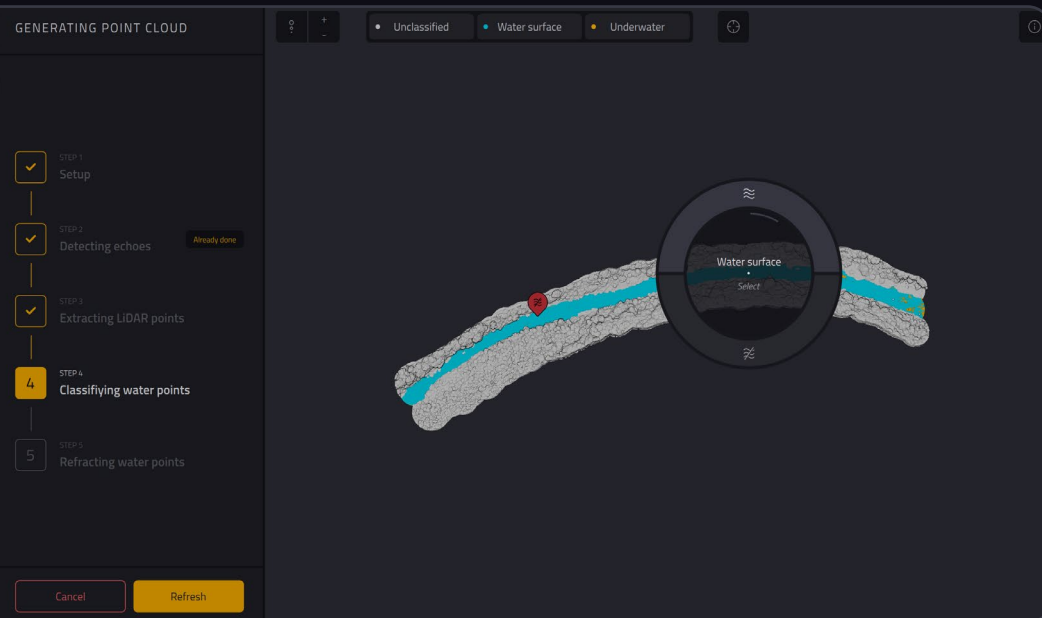
The echo detection can be a very lengthy process, depending on your flight conditions and MTA zone.

3

Extraction

Extracts the data a first time

Advanced Extraction Window.



► Navigator Advanced Extraction

4

Iterative Water Classification

First, an automated water surface detection that you can then refine. Picking a point in the point cloud opens a radial menu:

-  **Water Surface:** CloudStation will search for water surface around this location in the next iteration.
-  **Not Water Surface:** CloudStation will ignore any detected water surface around this location in the next iteration.
-  **Remove Constraint:** Available only when selecting an existing constraint. Select this dial to remove the constraint. This action cannot be undone.

Once satisfied, hit the Refresh button. You can repeat this process. If no constraint is added after an iteration, you can press Next.

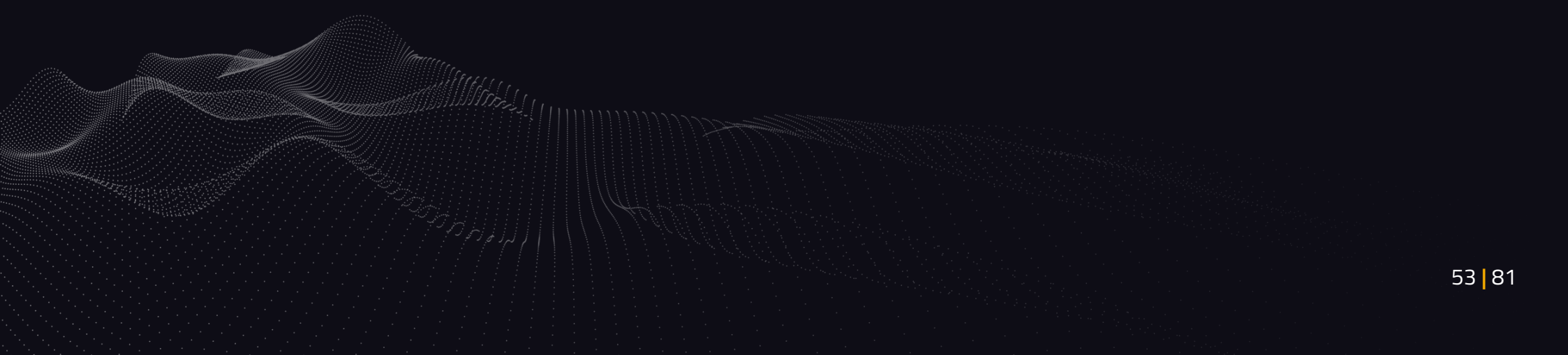
5

Water Points Refraction

All points under the defined water surface are refraction-corrected according to the value defined during the Setup step.

WORKFLOW

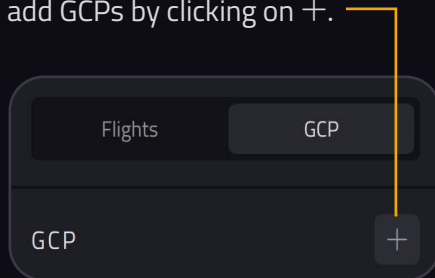
Data Refinement.



Ground Control Points. Pro

► Import & Display GCPs

Once extraction is complete, you can switch between Flights and GCP interface. You can add GCPs by clicking on +.



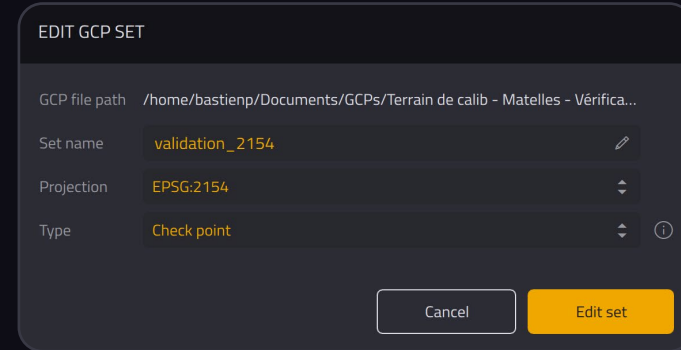
Your GCP file needs to comply with one of the following formats:

Comma separated CSV or TXT
Identifier, X, Y, Z

Space separated CSV or TXT
Identifier X Y Z

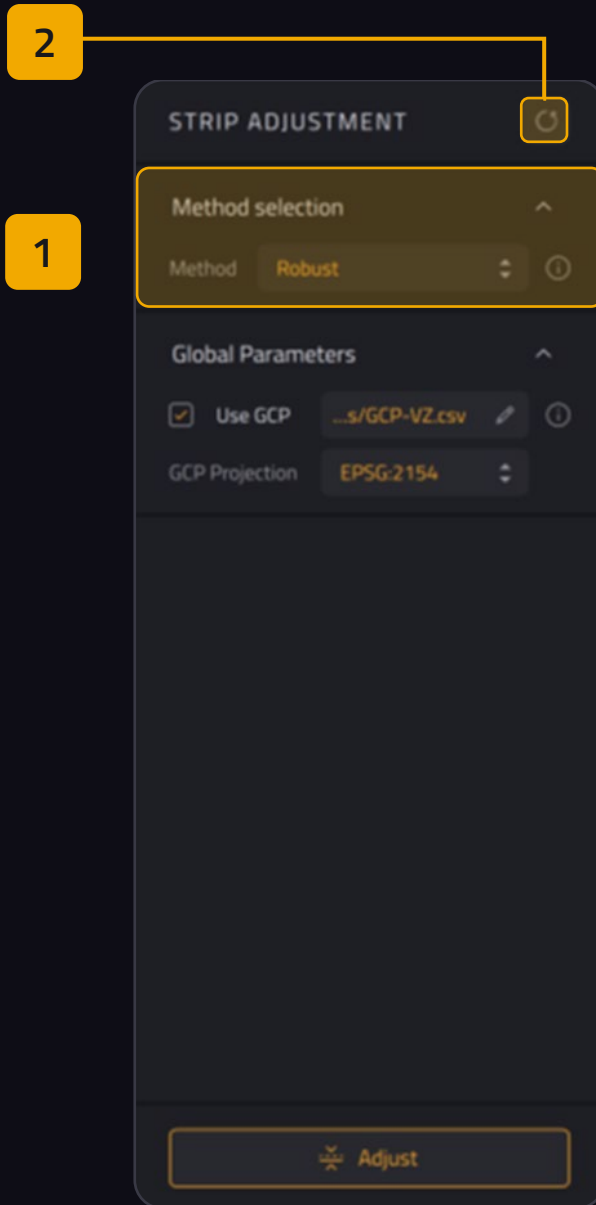
When providing GCPs, you will also need to select the projection used for them as well as what type of GCP.

Each set can be renamed just like flights. You can remove sets of GCPs or a specific point with .



CloudStation handles 3 different types of GCPs:

- ▲ **Tie Point:** Constraint point that can be used during the Strip Adjustment process.
- ◆ **Check Point:** Verification point. Not to be used in the Strip Adjustment process.
- **Point of Interest (POI):** Notable locations. Can be used to quickly navigate and identify locations in large datasets. These have no impact on data refinement.



Strip Adjustment. Pro

► Strip Adjustment Process

Quick Guide

Strip Adjustment module is meant to finely adjust the Strip-to-Strip misalignment that you may experience despite having your unit calibrated. It improves the quality of your data using 2 possible methods and can also use GCPs to adjust your data.

Prerequisites to Strip Adjustment: have more than one Strip with enough overlap (>25%) and flat areas large enough to be identified through multiple Strips.

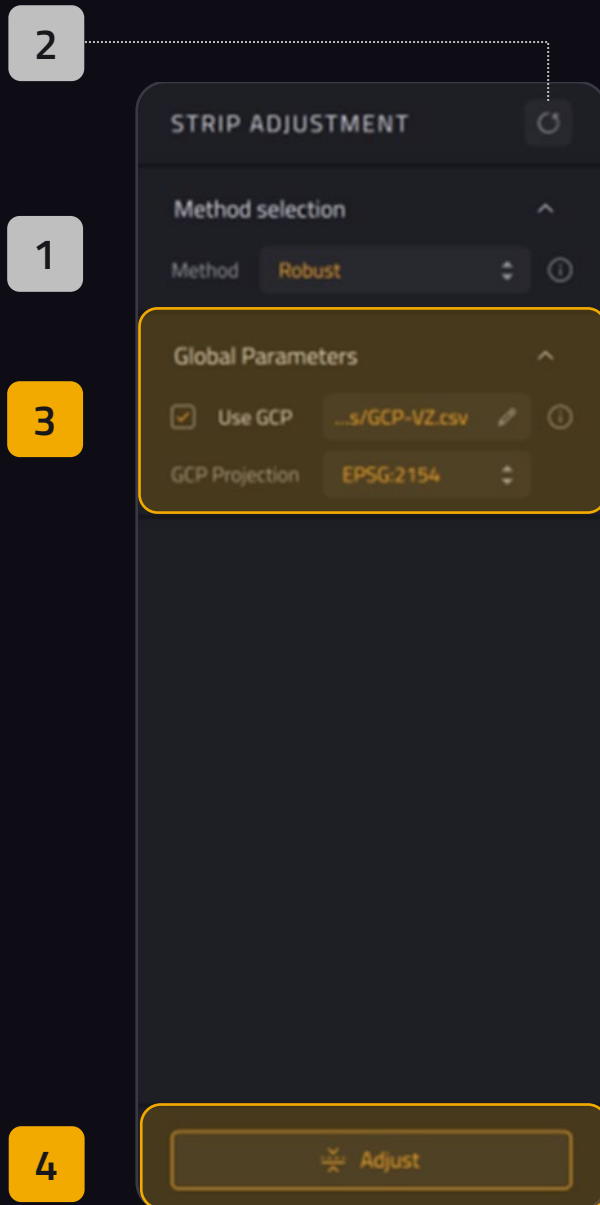
1 Method Selection

First, an automated water surface detection that you can then refine. Picking a point in the point cloud opens a radial menu:

- **Robust:** Default option, to use whenever you are not using GCPs and your Flight allows for multiple strips overlap
- **Precise:** Use this method for linear projects or when you are using GCPs.
- **Drive:** Only available with Fly & Drive-compatible scanners.

2 Reset to Default ↺

Go back to the default values for the Strip Adjustment algorithm.



Strip Adjustment. Pro

3 Ground Control Points

Though optional, you can include GCPs during Strip Adjustment processing.

The use of GCPs will result in a more accurate point cloud as the adjustment is done in parallel. *(More information: see next page)*

Your GCP file needs to comply with one of the following formats:

Comma separated CSV or TXT
Identifier, X, Y, Z

Space separated CSV or TXT
Identifier X Y Z

When providing GCPs, you will also need to select the projection used for them.



Collecting proper GCPs

GCP need to be collected on flat, hard surfaces with no vegetation above. Their layout and spread is key to obtaining great results with the Strip Adjustment algorithm. Having all GCPs collected only on one side of the surveyed area will impact the adjustment and give skewed results.

4 Main Button

Whenever you're ready to run the algorithm and all your settings are done, click the  Adjust button.

Strip Adjustment. Pro

► Influence of GCPs on Strip Adjustment methods

Strip Adjustment performs different actions depending on whether or not you have GCPs brought in. The table below summarizes the actions done.

Method	Best suited for...	GCP	Actions done
Robust	Area-type survey with short strips	No	Estimates boresight and a single translation correction per strip together with a per-beam (when relevant) calibration iteration.
		Yes	Estimates boresight and a single translation correction per-strip together with an intrinsic per-beam (when relevant) calibration iteration.
Precise	Corridor-like survey with longer strips and potential INS drift	No	Estimates time-dependent orientation and translation corrections together with a per-beam calibration iteration (if relevant).
		Yes	Estimates time-dependent orientation and translation corrections together with an intrinsic per-beam calibration iteration (if relevant).

Data Refinement. Pro

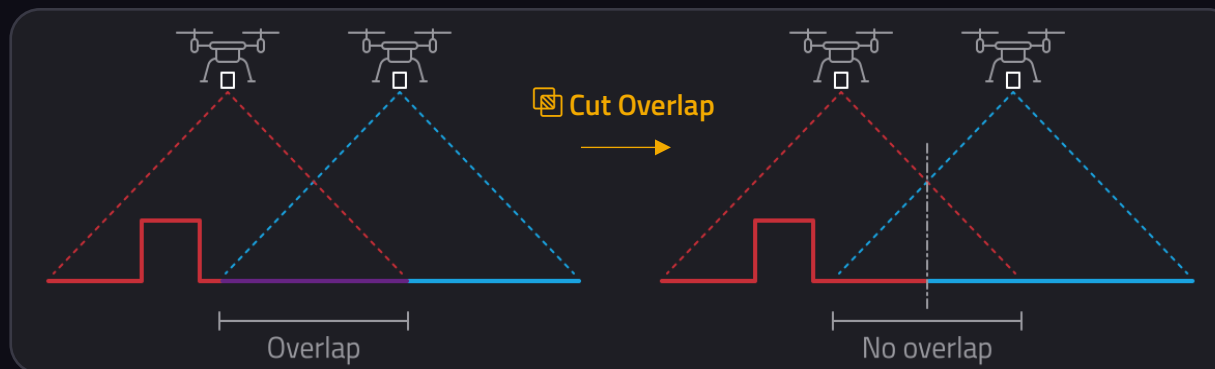
► Cut Overlap

Quick Guide

CloudStation offers a simple way to reduce unnecessary overlap in your data by smartly removing redundant data and keeping the higher quality points. This one-button operation is entirely automated.

When you are ready to process your data and remove overlap, simply hit the  **Cut Overlap** button.

The algorithm favors points closer to the LiDAR sensor as they are usually more accurate.



About Overlapping Points

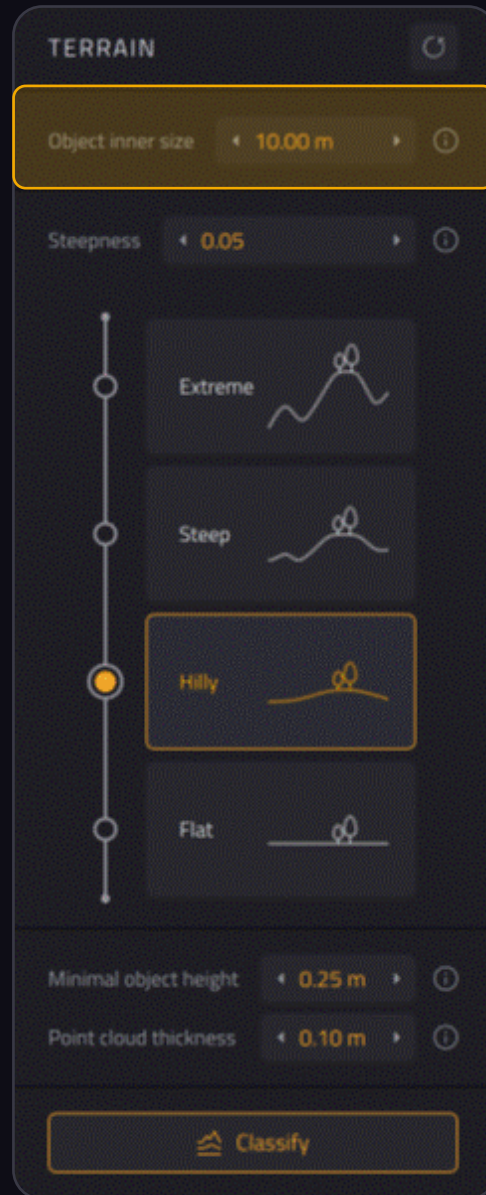
Points are not removed from the project, simply tagged and hidden.

WORKFLOW

Data Enrichment.

A solid yellow horizontal line positioned directly beneath the title 'Data Enrichment.'

1



Terrain Classification.

► Terrain Classification Process

Essential

Quick Guide

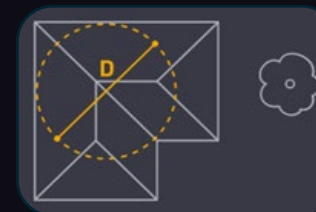
The Terrain module has two aspects to it. The first is ground classification of the point cloud, separating it from non-ground points. Once classified, the second part of the module allows the export of various digital models: DTM, DSM and DHM.

1

Object Inner Size

First, an automated water surface detection that you can then refine. Picking a point in the point cloud opens a radial menu: The maximal diameter of the largest part of non-ground object. The considered nonground objects can be man-made (buildings, bridges...) or natural (tree canopy, outcrop).

Use the measurement tool in the display panel to get a specific object diameter D .



1

2

3

TERRAIN

Object inner size

Steepness

Extreme

Steep

Hilly

Flat

Minimal object height

Point cloud thickness

Classify

Terrain Classification.

2

Steepness

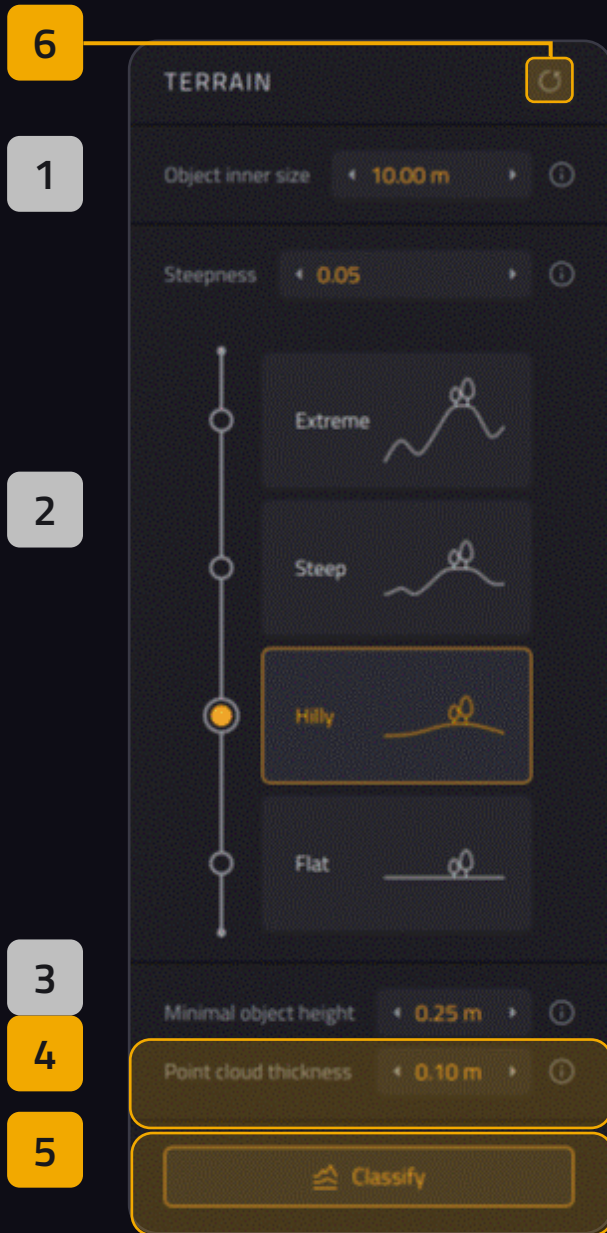
This value will reflect the terrain variations and how aggressively the filtering acts. The value ranges from 0 to 1. Preset values are illustrated as:

- **Extreme:** For rapidly changing terrain, with cliffs and flat parts. Typically, quarry terrain.
 - **Steep:** Mountainous type of terrain.
 - **Hilly:** Ground with regular variation. *This is the default value (0.05).*
 - **Flat:** Most aggressive filtering method for very flat terrain; Picture a parking lot.
- You can also manually enter a value if presets are not satisfactory.

3

Minimal Object Height

The vertical size above ground which an object is considered as not belonging to the ground (ie. a stump could be considered as or off-ground, depending on your final application).



Terrain Classification.

4

Point Cloud Thickness

The approximate thickness of your point cloud. This value is automatically filled depending on your system, but you can adjust it if you have noisier data or want to consider shrubs, for instance

5

Main Button

Whenever you're ready to run the classification and all your settings are done, click the  Classify button

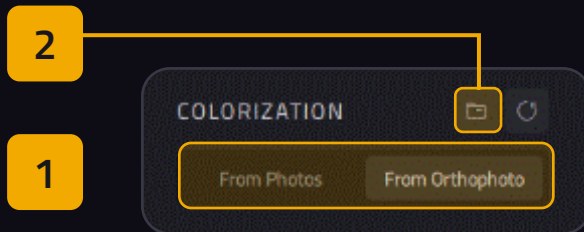
6

Reset to Default

Go back to the default values for the Terrain Module

Colorization.

► Colorization Options



1

Select Colorization Mode

Select the Colorization mode:

Pro

– **Color from Photos** : Use images from your Camera Module or a synchronized, external Camera to apply color to your point cloud. For more information about this Colorization method, see [Page 64](#).

Essential

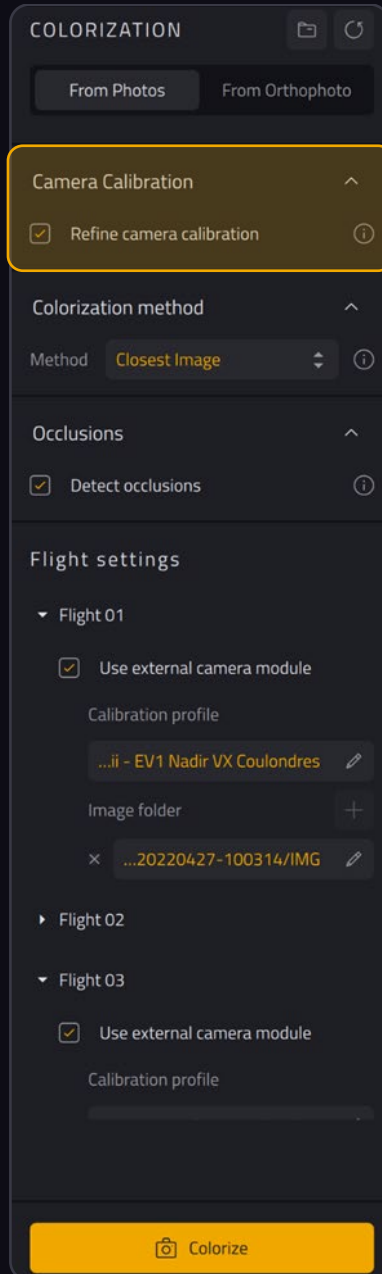
– **Color from Orthophoto**: Use a georeferenced orthophoto draped over your point cloud to apply RGB information. For more information about this Colorization method, see [Page 67](#).

2

Minimal Object Height

The vertical size above ground which an object is considered as not belonging to the ground (*ie. a stump could be considered as or off-ground, depending on your final application*).

1



Colorization.

► Color from Photos Pro

Quick Guide

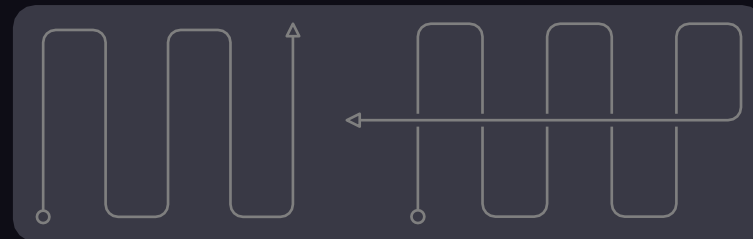
The Colorization module allows you to add RGB information to your LiDAR point cloud by using imagery acquired simultaneously through a camera module or an external camera. The creation of the PhotoID file prior to colorizing is now automated.

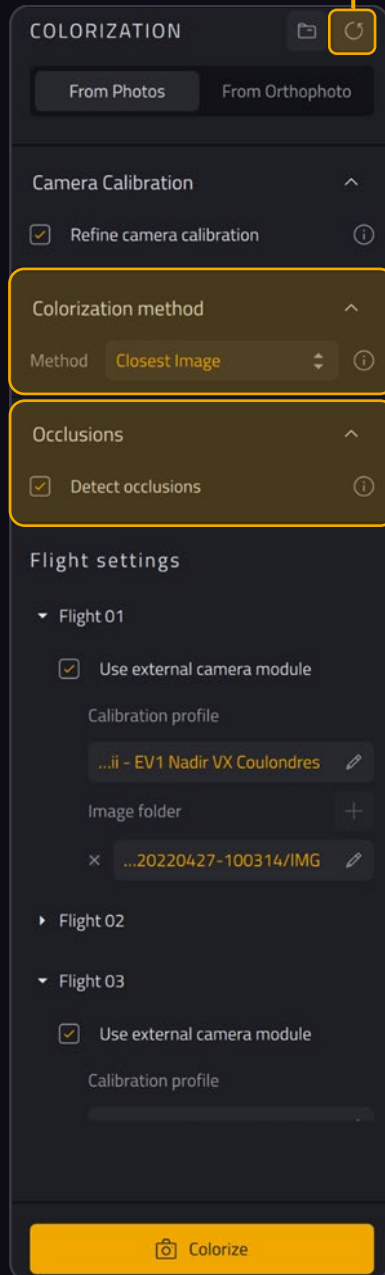
We recommend running  Strip Adjustment first whenever possible as the quality of the LiDAR point cloud will impact the colorization results.

1

Refine Camera Calibration

This option optimizes the camera calibration parameters for optimal data alignment before colorizing (recommended). If this option is not checked or should the calibration refinement fail, the factory camera calibration profile will be used for colorization. Make sure to add a transect to your Flights as the calibration refinement requires crossing Strips to properly run.





Colorization.

2

Colorization Method

Choose between two methods to apply RGB information to each point of the point cloud:

- **Closest Image:** Take the color information from the closest, non-occluded image available and applies the color to the point. Results in a faster and sharper colorization but seamlines may be more visible.
- **Median Color:** Consider the median RGB value from all non-occluded images and applies it to the point. Results in a smoother color with seamlines being blurred in, but the processing time is increased.

3

Occlusions

Uses the LiDAR point cloud to determine if any given point is visible from a photo or not. Color from photo(s) will only be considered if the point is visible when this option is ticked ON (recommended). The processing time increases significantly depending on the number of occlusions in the scene

4

Reset to Default ↺

Go back to the default values for the Colorization Module.

Warning: this will reset all the values, including the image directories.

4

1

2

3

5

6

COLORIZATION  

From Photos

From Orthophoto

Camera Calibration

☒ Refine camera calibration 

Colorization method

Method

Closest Image 



Occlusions

☒ Detect occlusions 

Flight settings

▼ Flight 01

☒ Use external camera module

Calibration profile

...ii - EV1 Nadir VX Coulondres 

Image folder

+

× ...20220427-100314/IMG 

▶ Flight 02

▼ Flight 03

☒ Use external camera module

Calibration profile

 Colorize

Colorization.

5

Flight Settings

This section has two different states. First, select for at least one Flight what type of images are provided between an Embedded Camera and a Camera Module.

Then, for at least one Flight, you will need to provide 3 things:

- **Location of the Image folder:** The images for this flight need to be saved to the same folder.
- **Camera:** Select the camera profile to use. The default value will be your favorite camera profile.
- **PhotoID File:** Generated automatically or from manual file creation, and placed in the \GNSS Post-Process Folder.

You cannot colorize without at least images from one flight. The Colorization algorithm runs on all flights. For flights without images, the ones from the other flights are used. *(A warning sign will then be displayed, but won't prevent you from processing).*

Images from the other flights are used to colorize this flight

▼ YS-20220427-111921 

6

Main Button

Whenever you're ready to colorize your point cloud and all your settings are done, click the  Colorize button.

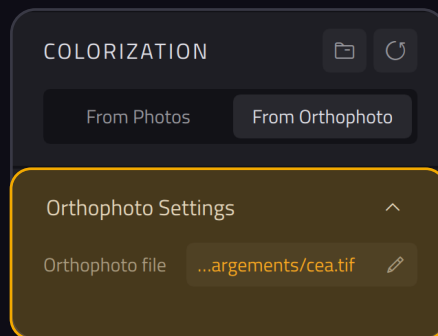
66 | 81

Colorization.

► Color from Orthophoto Essential

Quick Guide

The Color from Orthophoto allows you to drape an Orthophoto over your point cloud and attribute the RGB information to the points. Simply bring in your georeferenced Orthophoto to start the colorization process.



1

Orthophoto Settings

Select your Orthophoto file to be used for the Colorization process.

Accepted formats are:

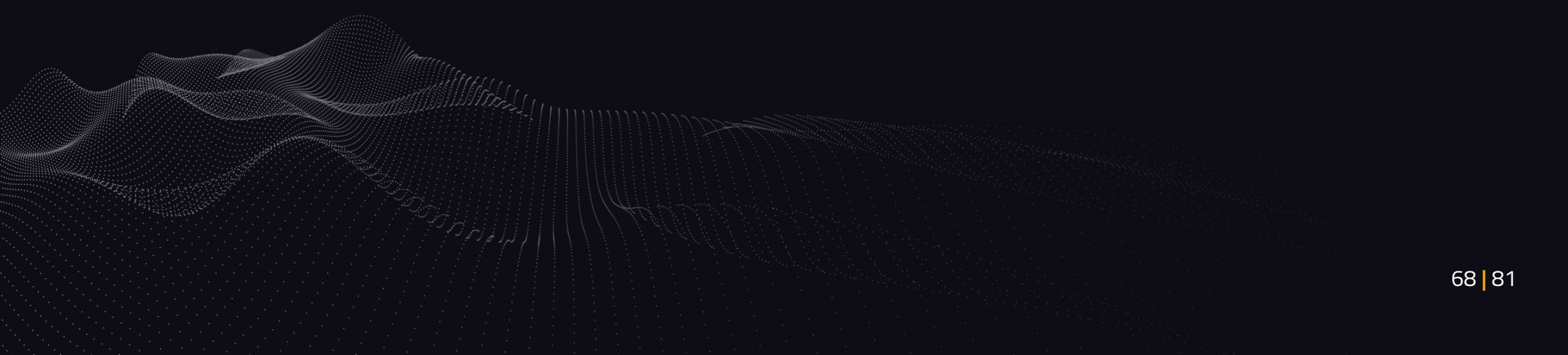
- *.TIF
- *.TIFF
- *.GEOTIF
- *.GEOTIFF



The Colorization result will directly be impacted by the quality and the accurate georeferencing of your file.

WORKFLOW

Data Export.



1

EXPORT

Point Cloud

Settings

Base name: Coulondres

Export as: LAS 1.4

Coordinates: EPSG:2154

Output folder: ...oup/export

Merging method: Entire project

Filters

Shape filter...: None

☒ Export point classes

☐ Export ground only

☐ Export withheld points

☒ Export colors

Select Strips

☒ Flight 01

1 2 3 4 5 6 7

☒ Flight 01

1

Export

2

Data Export.

Quick Guide

Multiple export types are available in CloudStation. When accessing the Export section, you will select what to export at the top of the Main Sidebar. Some types of exports can be restricted by your current licensing level.

► Data Export Selector

1

Export Type

Select the type of data you want to export.

Most of the Main Sidebar options will change accordingly.

Point Cloud

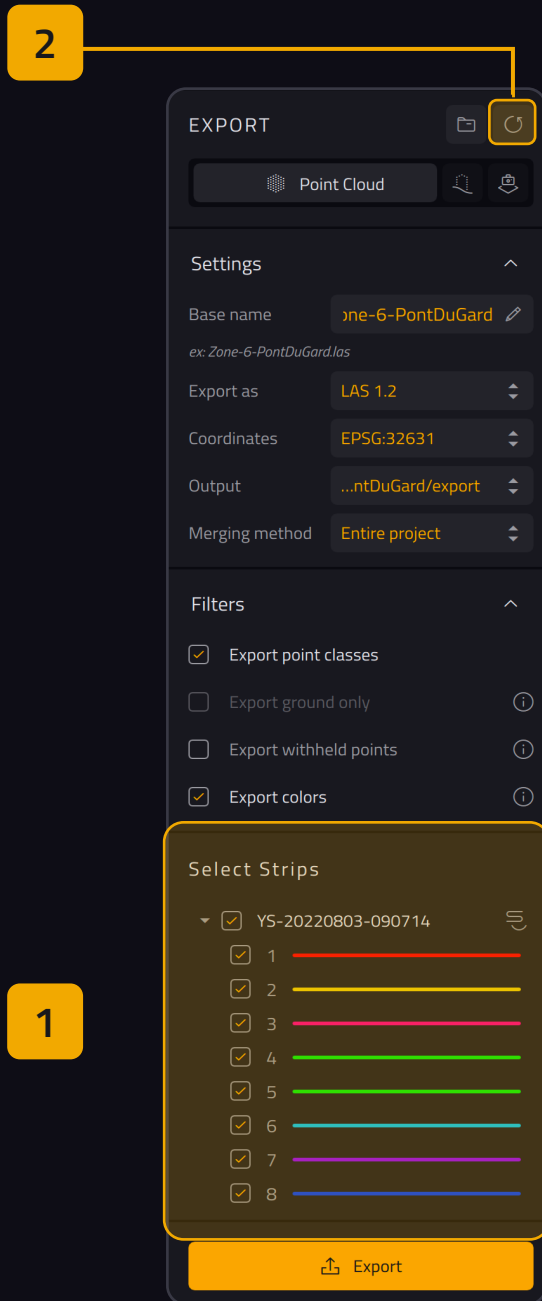
Digital Model

Orthophoto

2

Main Button

Whenever you're ready to export your data, hit the Export button. This is common to all types of exports.



Point Cloud Export. Essential

Quick Guide

Export your point cloud in the projection of your choice in a few clicks. You can merge everything or keep each Flight and Strip separate, and include classification and color information in your industry-standard *.LAS or *.LAZ file, or *.TXT.

► Point Cloud Export Parameters

1

Strip Selection

Select the Strips and Flights you want to export.

2

Reset to Default ↺

Reset to default your current export settings.

2

EXPORT

Point Cloud

Settings

Base name ex: Zone-6-PontDuGard.las

Export as

Coordinates

Output

Merging method

Filters

☒ Export point classes

☐ Export ground only

☐ Export withheld points

☒ Export colors

Select Strips

☒ YS-20220803-090714

☒ 1

☒ 2

☒ 3

☒ 4

☒ 5

☒ 6

☒ 7

☒ 8

Export

3

4

1

Point Cloud Export.

3

Settings

Set up your export parameters here:

- **Base name:** the common naming part for every file that will be created.
- **Export as:** Choose the format for your export between LAS 1.2 or 1.4, LAZ 1.2 or 1.4 and TXT (ASCII).
- **Coordinates:** Specify the coordinate system the data needs to be generated into. To learn more about custom coordinate systems, please refer to the next page.
- **Output:** Select the location where your data will be exported. If you have an ATIS.Cloud account, you can directly upload to ATIS.Cloud services.
- **Merging method:** By default, The whole project is merged as one file. You can also choose to merge them to have one file per Flight, or even per Strip.

4

Filters

Choose what to export here:

- **Export point classes:** include point classes in the data. All points are classified to the class 0 by default.
- **Export ground only:** only the points classified as ground will be exported.
- **Export withheld points:** When using CutOverlap, by default CloudStation will not export filtered points. You can force the export here.
- **Export colors:** adds RGB information to your exported data. Will default to elevation color values if no Colorization has been done..

Custom Coordinate System.

Quick Guide

CloudStation uses PROJ as the transformation software to convert geospatial coordinates from one reference system to another. It enables complex transformations from geodetic coordinates into local coordinate systems using pipeline operators.

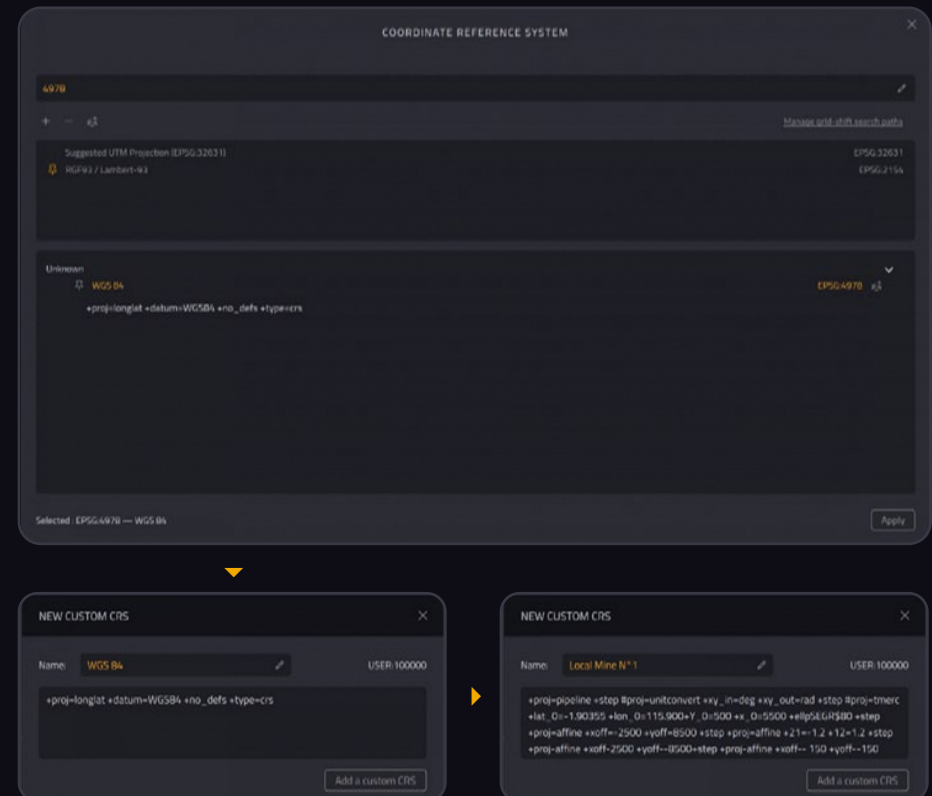
Further reading on PROJ software can be found at <https://proj.org/>.

► Create a Custom CRS

To create a new custom projection, open the Coordinate Reference System window by selecting « Other » in the dropdown menu. All custom projections have to start from EPSG:4978 (WGS84). Once highlighted, click on one of the two « Create a Custom projection based on this one » icons.

A new window prompt will allow you to name your new custom projection and enter the PROJ string to define it.

The projection is saved under the code USER:XXXXX. (*Incremental*)



1

EXPORT

Digital Model

Settings

Base name

Coulondres

ex: Coulondres_DSM.tif

Export as

DSM

DTM

DHM

Coordinates

EPSG:2154

Output folder

...oup/export

Common Properties

Pixel Size

0.50 m

Max Hole Size

20.00 m

DTM Properties

Z Value

Min Z

Mean Z

Max Z

Export hillshade

Type

Unidirectional

Multidirectional

Altitude

45 °

Export

Digital Model Export. Pro

Quick Guide

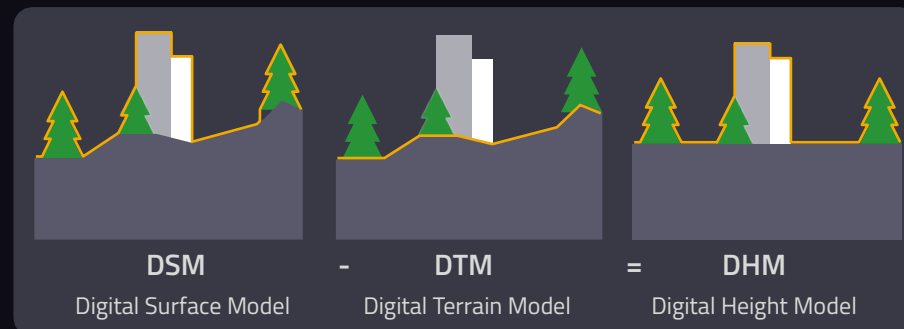
Once the Classification part of the Terrain module is complete, you can export multiple types of Digital Models by selecting the corresponding Export.

Digital Model Export Parameters

1 Settings

This section allows you to choose a base name for your export, determine the Coordinate System used as well as pointing where to save each exported Digital Model. Choose what type of Digital Model you wish to export:

- **DSM (Digital Surface Model):** Export a raster of the surveyed area containing both the ground and off-ground objects. Available without Terrain Module.
- **DTM (Digital Terrain Model):** Export a raster of the ground level only, based on the points plus a gap filling value.
- **DHM (Digital Height Model):** Export a raster of the surveyed area based on the difference between the DSM and the DTM.



The screenshot shows the 'EXPORT' panel for a Digital Model. It includes a 'Settings' section with fields for 'Base name' (Coulondres), 'Export as' (DSM, DTM, DHM), 'Coordinates' (EPSG:2154), and 'Output folder' (...oup/export). Below this are four expandable sections: 'Common Properties' (Pixel Size: 0.50 m, Max Hole Size: 20.00 m), 'DTM Properties' (Z Value: Min Z, Mean Z, Max Z), 'Hillshade Properties' (Export hillshade: checked, Type: Unidirectional/Multidirectional, Altitude: 45°), and a 'Reset to Default' button. A yellow line connects callout 5 to the 'Reset to Default' button.

Digital Model Export.

2 Common Properties

Adjust these values regardless of the type of Digital Model you wish to export.

- **Pixel Size:** Corresponds to the distance considered in the area covered by a pixel. A smaller pixel size value will result in a larger raster file but will also give a finer description of the terrain. However, a Pixel Size too low may be more sensitive to noise and data gaps.
- **Max Hole Size:** Set the largest data gap you wish to fill in the Digital Model.

3 DTM Properties

This option is only available when exporting a DTM.

- Choose the Z value attributed to a pixel between the lowest point, the highest point or the mean Z value in the area.

4 Hillshade Properties

Adjust these values regardless of the type of Digital Model you wish to export.

- **Pixel Size:** Corresponds to the distance considered in the area covered by a pixel. A smaller pixel size value will result in a larger raster file but will also give a finer description of the terrain. However, a Pixel Size too low may be more sensitive to noise and data gaps.
- **Max Hole Size:** Set the largest data gap you wish to fill in the Digital Model.

5 Reset to Default

Reset to default your current export settings.

EXPORT

Orthophoto

Settings

Base name: **Coulondres**
ex: Coulondres_orthophoto.tif

Coordinates: **EPSG:2154**

Output folder: **...oup/export**

Properties

Strategy: **Fixed**

Pixel Size: **0.02 m**

Flight settings

▼ Flight 01

☒ Use external camera module

Calibration profile
...li - EV1 Nadir VX Coulondres

Image folder
...20220427-100314/IMG

► Flight 02

► Flight 03

Export

Orthophoto Export. Ultimate

Quick Guide

CloudStation can generate orthophotos from simultaneously-acquired images, without requiring as much overlap as standard photogrammetry processes as the LiDAR data is considered and helps with the generation. Choose your pixel size and coordinates settings and hit Export!

► Orthophoto Parameters

2

Settings

This section allows you to choose a base name for your export, determine the Coordinate System used as well as pointing where to save your Orthophoto.

3

Flight Settings

Similar to the Color from Photos flight settings, choose which type of camera you'll be using first, then point the image folder as well as which calibration profile to use.



While it is possible to create orthophotos from embedded cameras from our systems, we do not recommend it as they're not meant for it.

1

2

3

EXPORT

Orthophoto

Settings

Base name

Coulondres

ex: Coulondres_orthophoto.tif

Coordinates

EPSG:2154

Output folder

...oup/export

Properties

Strategy

Fixed

Pixel Size

0.02 m

Flight settings

Flight 01

Use external camera module

Calibration profile

...li - EV1 Nadir VX Coulondres

Image folder

...20220427-100314/IMG

Flight 02

Flight 03

Export

Orthophoto Export. Ultimate

3

Properties

Two different strategies are available for Orthophoto export:

Fixed

Input the Pixel Size value you want to obtain in your final product.

Multiple of GSD

Input the average flight height for your mission for CloudStation to determine the GSD for your project, then a multiple of the GSD you want for your orthophoto.



The orthophoto process generates your ortho at the same time as exporting it. It will take longer than a standard export.

WORKFLOW

Troubleshooting.

A solid yellow horizontal line underlining the word 'Troubleshooting.'

General Issues.

► Installation and missing DLL files

If you get an error message at start, stating that VCOMP140.DLL is missing, please download the **Visual C++ Redistributable for Visual Studio 2015** [here](#).

► CloudStation won't start

If CloudStation doesn't start, make sure to try the following steps:

- Uninstall and reinstall CloudStation
- Restart your computer
- Update your GPU driver and restart your computer

For **Nvidia** drivers, you can look for your GPU [here](#).

For **AMD** drivers, you can look for your GPU [here](#).

(Note: we do not recommend using AMD GPUs for CloudStation)

If you are still experiencing issues, please contact Customer Support.

► Access to Log Files

Each CloudStation session records a log file that can be used to troubleshoot abnormal software behavior or crashes.

To access the log files, go to the user local app data. To do so, open a Windows Explorer and enter:

%localappdata%\YellowScan\CloudStation\logs

or directly navigate to :
C:\Users\[your_user_name]\AppData\local\YellowScan\CloudStation\logs\

You can also find an Export Logs button at the bottom of the General section of the Settings. This button will bundle up the last 10 log files for you to easily share with the Support Team.

Support

Support team can ask you to export your logs. Click on the button below to export your latest logs.

Export logs

License Troubleshooting.

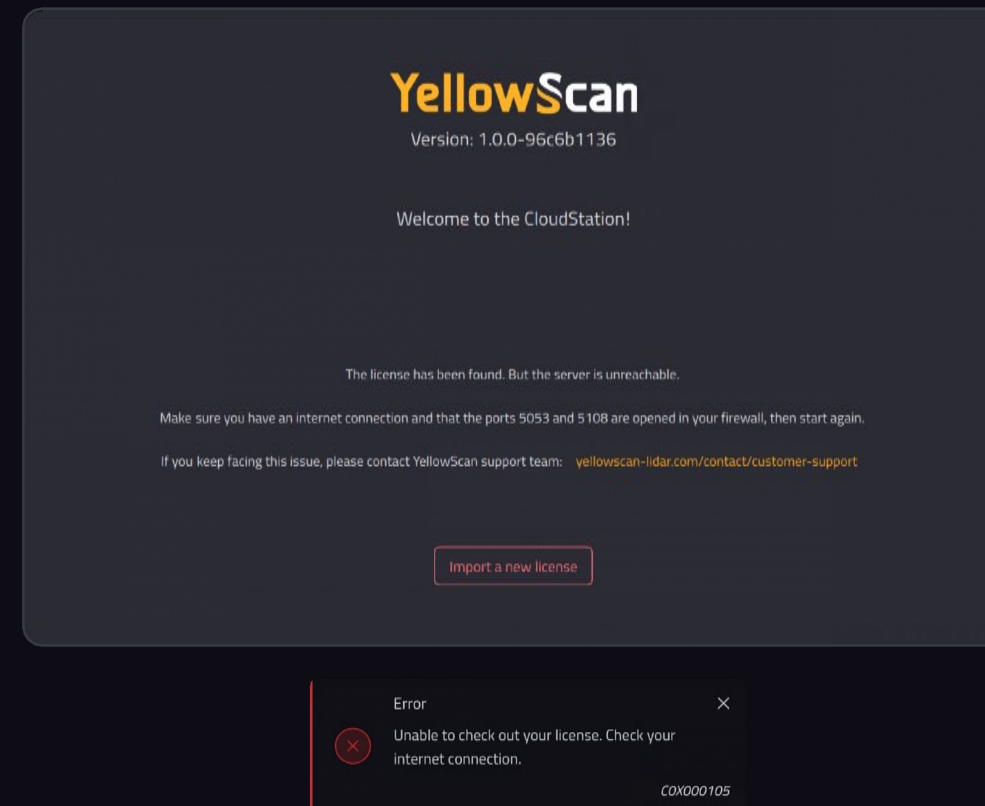
► License Server Access

When the import of a new license or opening CloudStation with a valid license does not work:

- Check your Internet connection. You must have a valid internet connection when importing a new license or opening CloudStation if offline mode is not enabled (and has not expired).
- Some ports must be opened in all the firewalls that are between your machine and the internet (this includes Windows Firewall, your Company firewall, etc.).

These ports are:

- TCP out 5053
- TCP out custom. CloudStation will inform you of this port number on the first connection attempt if it fails.
- Check that another computer is not already using your license (to make a license available again, close CloudStation on the other device or get out of offline mode).
- Your device and all the devices it passes through (such as a phone hotspot or some routers) must have IPv4 support enabled. Our license server is not compatible with IPv6-only connections



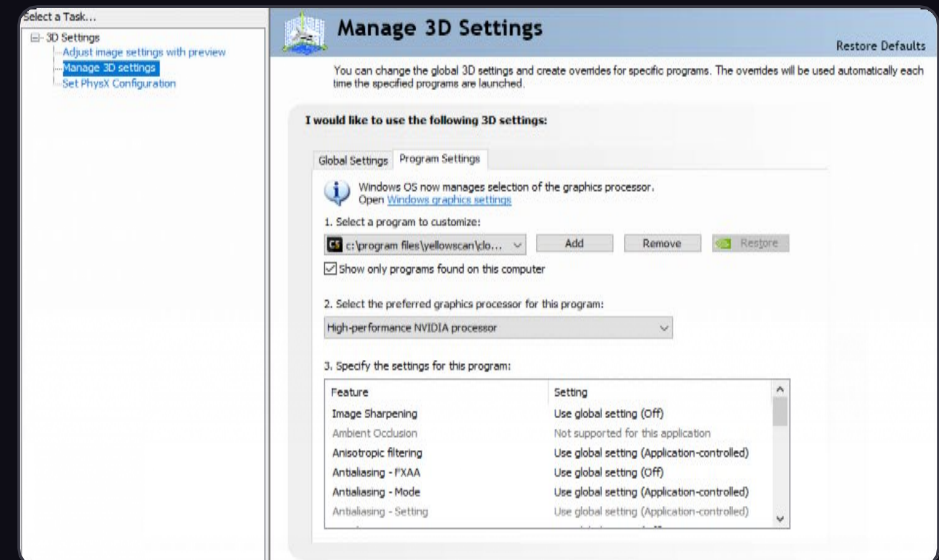
Performance Troubleshooting.

► Force Use a Dedicated High Performance GPU

Some laptops have two GPUs: an integrated one (energy efficient) and a dedicated one (for applications that require more power). CloudStation can use your GPU for two purposes:

- Computing: go to Settings > System and enable OpenCL, then check your most powerful GPU. This will speed up some algorithms in CloudStation
- Visualization: displaying the point cloud

To force the use of the dedicated high-performance GPU, open the Nvidia Control Panel and navigate to Manage 3D Settings > Program Settings. Select CloudStation.exe in the list of applications to enforce the use of your Nvidia GPU.





Contact Support.

- ▶ support@yellowscan.com
- ▶ +33 (0)4 11 93 14 14
- ▶ www.yellowscan.com/contact#support